

UNIVERSITY OF OTAGO

DEPARTMENT OF COMPUTER SCIENCE

SCNC597 PROJECT REPORT

**A Comparative Study of Deep Learning Architectures for
Representation Learning from Ocular Fundus Images**

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Contents

ABSTRACT.....	3
1. INTRODUCTION.....	3
2. RELATED WORK.....	4
2.1 SELF-SUPERVISED AND FOUNDATION MODELS FOR RETINAL IMAGING.....	4
2.2 MECHANISMS FOR REPRESENTATION LEARNING.....	5
2.3 TRANSFER LABEL EFFICIENCY UNDER DATA SCARCITY.....	5
2.4 RELATION TO PRESENT STUDY.....	6
3. DATA PREPROCESSING	6
3.1 IMAGE RESIZING AND CROPPING	7
3.2 LOADING -TIME TRANSFORMS.....	7
3.3 DATASET ASSEMBLY AND HELD-OUT SPLIT.....	8
4. METHODOLOGY.....	9
4.1 CONVOLUTIONAL AUTOENCODER (CAE)	10
4.1.1 <i>No-Skip</i>	11
4.1.2 <i>Two Early Skips</i>	12
4.1.3 <i>Two deep Skips</i>	12
4.1.4 <i>All Five Skips</i>	12
4.2 DIFFUSION-BASED MODEL	13
4.2.1 <i>No-Skip</i>	13
4.2.2 <i>Two early Skips</i>	13
4.2.3 <i>All Five skips</i>	14
4.3 GRAPH BASED AUTOENCODER (GAE).....	14
5. EVALUATION	16
5.1 FROZEN REPRESENTATIONS	16
5.1.1 <i>Linear Probe</i>	16
5.1.2 <i>Label-Purity</i>	17
5.1.3 <i>Raw-Pixel Baseline</i>	17
5.1.4 <i>Cluster Separation</i>	17
5.1.5 <i>Embedding Visualization</i>	18
5.2 FINE-TUNING	18
5.2.1 <i>Models Selected</i>	18
5.2.2 <i>Trainable Components</i>	18
5.2.3 <i>Training Data and Label Budgets</i>	19
5.2.4 <i>Optimisation</i>	19
5.2.5 <i>Evaluation</i>	19
6. RESULTS	20
6.1 RESULTS OF FROZEN REPRESENTATIONS.....	20
6.2 FINE-TUNING	22
7. ANALYSIS.....	24
7.1 COMPARISON AGAINST THE RAW-PIXEL BASELINE	24
7.2 EFFECT OF THE ENCODER MECHANISM	24
7.3 EFFECT OF SKIP CONNECTIONS.....	25

7.4 THE TWO HIGHEST-SCORING DIFFUSION MODELS.....	26
7.5 LABEL PURITY.....	26
7.6 SECONDARY MEASURES AND AN ANOMALY.....	26
7.7 SUMMARY OF THE FROZEN REPRESENTATION	27
7.8 OVERALL PATTERN	27
7.9 GRAPH AUTOENCODER	27
7.10 DIFFUSION MODELS	28
7.11 EFFECT OF LABEL BUDGET	28
7.12 SUMMARY OF FINE-TUNING	29
8. CONCLUSION.....	29
REFERENCES.....	31
APPENDIX A.....	32
A1. EMBEDDING PROJECTIONS.....	32
A2. LEARNING CURVES	35
APPENDIX B.....	38
B1. ARCHITECTURE DIAGRAM FOR MODELS DURING EVALUATION.....	38
B2. ARCHITECTURE DIAGRAM DURING FINETUNING	39
STATEMENT OF AI USE	41

Abstract

Learning useful features from unlabelled retinal images depends on the type of encoder used, but convolutional, diffusion, and graph-based encoders are rarely compared under the same conditions. This study trains all three as image autoencoders with the same dataset, latent size, training objective, and training budget, holding out age-related macular degeneration as an unseen test disease, and measures how well their frozen representations separate it from a raw-pixel baseline. Only the diffusion models with skip connections and the graph-based autoencoder scored above the baseline. When the leading models were fine-tuned with small numbers of labels, the graph-based autoencoder improved and passed its own frozen result, while both diffusion models stayed below theirs at every budget. The encoder mechanism, rather than the use of unsupervised reconstruction alone, determined whether a representation separated the unseen disease, and the strongest frozen representation was not the one that adapted best to fine-tuning.

1. Introduction

The retina can be imaged directly and without surgery, and it shares tissue and blood supply with the brain. For this reason, it has been described as a window to the brain [8], and there is growing interest in whether retinal images carry signs of conditions that affect the central nervous system, including dementia. If such signs can be detected from a fundus photograph, retinal imaging could offer a low-cost and non-invasive way to support screening.

This project began with the intention of working with dementia-labelled fundus images. The original plan was to use data from a small-scale study based in Dunedin, and additional efforts were made to obtain suitable data from the UK Biobank. Neither source became available within the scope of this work. Because no dementia-labelled fundus dataset could be used, dementia could not serve as the target condition directly.

Age-related macular degeneration (ARMD) was therefore used as a stand-in target. ARMD is a retinal disease for which labelled images are available in public datasets, which makes it possible to test whether a representation learned without disease labels can separate a disease it was never trained on. ARMD is not the clinical condition of interest, and no claim about dementia is made from the results. It is used because it allows the central question to be examined under realistic conditions of limited labelled data, which is the setting any later work on dementia would face.

That central question concerns the encoder. Learning useful features from unlabelled retinal images depends on the kind of network used to encode them, but the encoder types available for this are rarely compared directly under the same conditions. This

study compares three encoder mechanisms — a convolutional autoencoder, a diffusion-based denoising model, and a graph-based autoencoder — as image autoencoders that share the same dataset, latent size, training objective, and training budget, so that the encoder mechanism is the only thing that changes between them. All are trained from scratch, with ARMD held out from training, and their frozen representations are evaluated on how well they separate this unseen disease. A second experiment measures how the leading models behave when a small number of labelled examples is used to fine-tune them, since label efficiency is central to any eventual application where labelled data is scarce. The aim of this study is to determine which encoder mechanism produces the most transferable frozen representation of an unseen disease, and whether that ranking holds when the representations are fine-tuned with a small number of labels.

2. Related Work

This study draws on three areas of prior work: self-supervised and foundation models for retinal imaging, the use of different neural network types as representation learners, and transfer and label efficiency when labelled data is limited. Each area is reviewed below.

Self-Supervised pretraining is learning from images without diagnostic labels. It does this by training the model on a task built from the images itself, such as reconstructing or removing added noise, which requires it to capture the structure of the images. The internal features it forms can be reused for later labelled tasks.

2.1 Self-Supervised and Foundation Models for Retinal Imaging

A common recent approach to learning from unlabelled fundus images is large-scale self-supervised pretraining. RETFound [1] trained a masked-autoencoder Vision Transformer on about 1.6 million unlabelled retinal images and then adapted it to several downstream tasks using few labels. Later work has asked whether this scale is needed. A retinal model trained on about 75,000 public fundus images, using far less compute, reported similar downstream performance [2], which suggests that useful retinal representations can be learned at a smaller scale. Other models add text or multimodal supervision rather than using images alone [3].

These representations do not always transfer well. Silva-Rodríguez et al. [13] found that when features from self-supervised retinal pretraining were frozen and only a simple linear classifier was trained on top, transfer to new tasks was limited, and in some cases performed worse than a simple ImageNet-pretrained baseline. They noted that these models often need full finetuning to adapt well. This is relevant to this present study, because it shows that the quality of frozen retinal representations is still an open question.

This study differs from the foundation-model work reviewed above [1]-[3] in three ways. First, instead of fixing one architecture and increasing the amount of data, it keeps the

dataset, latent size, optimiser, and training budget the same and changes only the encoder type, so that any difference in results can be linked to the encoder rather than to data volume or model size. Second, all models are trained from scratch, without ImageNet or foundation-model initialisation. Because the study compares encoder mechanisms, any transfer ability inherited from outside datasets would confound that comparison. Training from scratch ensures that transfer to the held-out disease reflects what the encoder learned from the retinal data in this study. Third, it tests transfer to a disease that is removed from training completely, rather than to tasks drawn from the training distribution.

2.2 Mechanisms for Representation Learning

The three encoder types compared in this study have each been used before as representation learners. Convolutional autoencoders have been used to extract features from medical images for later analysis [4], learning compact representations through pixel-level reconstruction. Diffusion and denoising models have more recently been shown to learn representations useful beyond image generation. Mittal et al. [5] adapted the denoising objective specifically for representation learning, and several studies extract features from the bottleneck of a denoising network at a fixed noise level for classification, which is the approach used here. The denoising diffusion probabilistic model [6] is the basis for the diffusion variants in this study.

Graph-based encoders are the third type. The dynamic graph convolution of Wang et al. [7] rebuilds a k -nearest-neighbour graph in feature space at each layer, so information can pass between regions that are similar in feature space but not next to each other in the image. A standard convolution cannot do this in a single step. This study applies that idea to image patches, using similarity-based message passing as the encoder of an image autoencoder.

These encoder types are rarely compared directly under the same conditions; most studies look at one type on its own. This is the reason for the controlled comparison used here, in which the encoder type is the only thing that changes between models.

2.3 Transfer Label Efficiency Under Data Scarcity

This study is motivated by the limited availability of dementia-labelled fundus data, which makes the data efficiency of learned representations important. The retina can be imaged directly and non-invasively and is connected to the central nervous system [8], and retinal measures have been linked to cognitive decline and dementia risk [9]; this is why fundus photography is of interest as a screening method. However, dementia-labelled fundus datasets are scarce. Self-supervised pretraining is one known way to reduce the amount of labelled data needed in medical imaging, and the foundation-model results above are examples of this.

This study asks not only whether transfer happens, but which encoder type transfers best to a disease not seen during training, and how performance changes as the number of

labelled examples is reduced. This label-efficiency analysis is relevant to the eventual dementia setting, where only small labelled datasets are likely to be available.

2.4 Relation to Present Study

Taken together, the work reviewed above leaves a clear gap. Foundation models show that retinal representations can be learned without labels, but they use one architecture and change the scale, so the effect of the encoder type cannot be separated from the effects of data volume and model size; the evidence on frozen-representation transfer is also mixed rather than settled [1–3]. The mechanism work shows that convolutional, diffusion, and graph encoders are each able to learn useful representations [4–7], but rarely compares them under the same conditions. The scarcity motivation [8, 9] raises the question of label efficiency without showing which encoder type transfers best when labels are few.

This study addresses that gap. It implements convolutional, diffusion, and graph-based encoders as image autoencoders that share the same training setup: the same dataset, latent size, reconstruction objective, optimiser, and training budget, so that the encoder type is the only variable that changes. All models are trained from scratch, with the target disease (age-related macular degeneration) removed from training, and their frozen representations are evaluated on how well they separate this unseen disease. The aim is to find which encoder type produces the most transferable representations under these conditions, and to measure how well that transfer holds as the amount of labelled data is reduced. This is intended to provide a methodological basis for the later application to dementia, where labelled fundus data is scarce and a data-efficient, transferable representation is likely to be important.

3. Data Preprocessing

Preprocessing was carried out in two stages. The first stage operated on the image files and produced a single fixed-size version of every image. The second stage was at data-loading time during training and evaluation, and applied tensor conversion, optional augmentation, dataset assembly, and the held-out split used for the downstream task. The two stages are described in turn. The dataset splits were not created during preprocessing. Each of the three datasets was obtained from Kaggle with its training, validation, and test divisions already provided. These divisions are used only for the unsupervised representation training, which uses no labels, so the appearance of the same disease types across divisions does not introduce label leakage. The separate evaluation on the held-out disease, described in Section 3.3, is the split that isolates a disease not seen during training: ARMD-positive images were excluded from representation training and used only for downstream evaluation.

3.1 Image Resizing and Cropping

All images were first passed through a resizing step that standardised them to a common size. Fundus photographs contain a circular retinal region surrounded by a black border, and the extent of this border varies between images and datasets. To reduce this variation, each image was first cropped to the region containing the retina. The image was converted to greyscale and thresholded at a fixed intensity value to separate the bright retinal area from the dark background. Morphological closing and opening were then applied to the resulting mask to remove small gaps and isolated noise, and the bounding box of the remaining non-zero region was taken as the crop area. A small fixed padding was added to each side of this bounding box, with the padding clamped to the image boundaries so that it could not extend beyond the original image. Figure 1 shows an example fundus image before this processing and Figure 2 shows the same image after cropping and resizing.

After cropping, each image was resized to 512 by 512 pixels. This resize does not preserve the aspect ratio of the cropped region, so non-square crops are stretched. Measured across all images, the crop aspect ratio (width/height) had a median of 1.00 and a mean of 1.045, with 80% of images falling between 0.9 and 1.1. Most crops were close to square and were only barely distorted by the square resize. A small number of images had extreme ratios, with a maximum of 13.3, reflecting cases where border removal produced a highly non-square region; these are a limitation of the fixed-threshold cropping. The same resize was applied to every image and every model, so any distortion affects all models equally and does not bias the comparison between them.

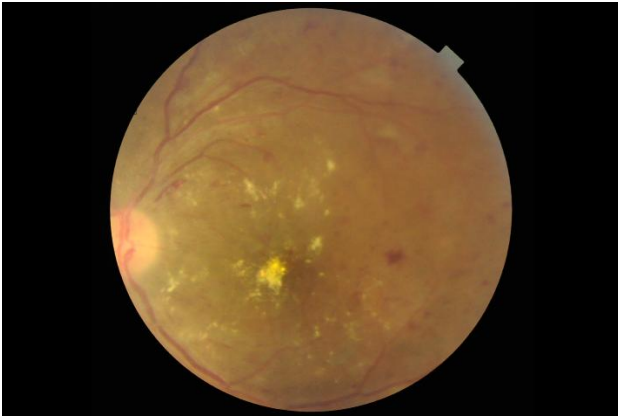


Figure 1. Original (2144x1424)

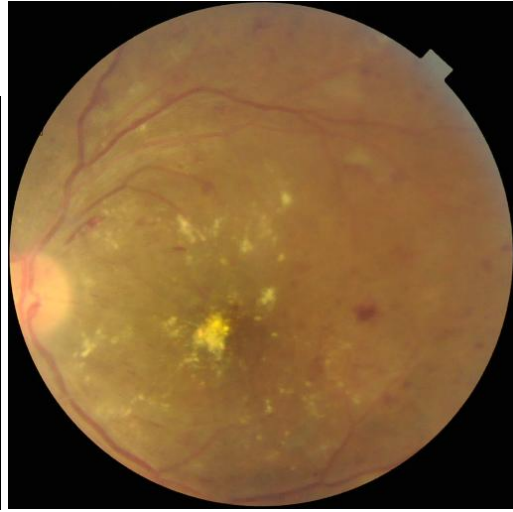


Figure 2. Post Processing (512x512)

3.2 Loading -Time Transforms

During training and evaluation, images were read from the resized directory and converted to tensors. Conversion scaled the pixel values from the original 0–255 integer

range to the 0–1 floating-point range. No mean or standard-deviation normalisation was applied. The 0–1 range was kept because the autoencoder decoders end in a sigmoid activation, whose output also lies in the 0–1 range, so the input and the reconstruction target occupy the same range.

Augmentation was applied only to data drawn for training. When enabled, it consisted of a random horizontal flip and a random vertical flip; no other augmentation, such as rotation, colour adjustment, or cropping, was used. Validation and evaluation data were converted to tensors without augmentation.

3.3 Dataset Assembly and Held-out Split

The wider motivation for this work concerns dementia, where retinal imaging has been suggested as a possible non-invasive source of biomarkers. Fundus datasets with dementia labels were not available for this project, so dementia could not be used as the target condition directly. Age-related macular degeneration (ARMD) was used in its place as the held-out disease. ARMD is a retinal condition for which labelled images are available within the datasets used here, which makes it possible to test whether a representation learned without those labels can separate an unseen disease. ARMD therefore serves as a stand-in target that allows the transfer question to be examined under realistic label scarcity; it is not itself the clinical condition of interest, and no claim about dementia is made from these results.

Three datasets were used for representation learning: EyePACS [10], RFMid [11], and ORIGA [12]. EyePACS is a large diabetic-retinopathy screening dataset and contributed 92,501 images, divided into 73,995 training, 9,245 validation, and 9,261 test images. RFMid is a multi-disease retinal dataset with labels for 45 conditions and contributed 3,200 images, divided into 1,920 training, 640 validation, and 640 test images. ORIGA is a glaucoma dataset and contributed 650 images, divided into 454 training and 196 test images; it has no validation split. In total, 96,351 images were used. The three datasets were combined into a single training source, with images loaded from the existing split folders of each dataset. The validation source did not include ORIGA, which has no validation split.

The held-out evaluation target, ARMD, is recorded as a label within RFMid. Across the RFMid splits there are 169 ARMD-positive images, with 100 in the training split, 38 in the validation split, and 31 in the test split. Any RFMid image labelled ARMD positive was excluded when loading data for representation learning, so that no ARMD-positive image was seen during training. The remaining RFMid images, together with all EyePACS and ORIGA images, formed the training data. Examples of ARMD-positive and non-ARMD fundus images are shown in Figure 3 and 4.



Figure 3. ARMD Positive sample



Figure 4. Non-ARMD sample.

For the downstream evaluation, a separate data source was built from the RFMid images only, in which each image was labelled according to whether it was ARMD positive or not. From this source a fixed test split was constructed by selecting a set number of ARMD-positive images and an equal number of non-ARMD images, giving a balanced test set. The selection used a fixed random seed so that the same test split was produced on every run and was identical across all models. The images not placed in the test set formed the pool used for probe training and for the label-limited fine-tuning experiments. Because the split depends only on the fixed seed and the dataset contents, it is reproducible and consistent across the different models being compared.

4. Methodology

This study compares three encoder mechanisms for unsupervised representation learning from retinal fundus images: a convolutional autoencoder (CAE), a diffusion-based denoising model, and a graph-based autoencoder (GAE). All three are implemented as image autoencoders with the same input and output: a fundus image is encoded into a spatial latent representation and decoded back to a reconstruction of the same image, trained with an L1 reconstruction loss. The dataset, latent channel dimensionality, optimiser, learning-rate schedule, and training budget are held constant across all models, so that any difference in the learned representations can be attributed to the encoder mechanism rather than to other factors. A 512-dimensional embedding is obtained from each trained model by spatial mean-pooling of its latent, and this embedding is used for all downstream evaluation with the encoder weights frozen.

A spatial latent is used rather than a single flattened bottleneck vector. A flattened bottleneck forces all information through one narrow channel, which is particularly

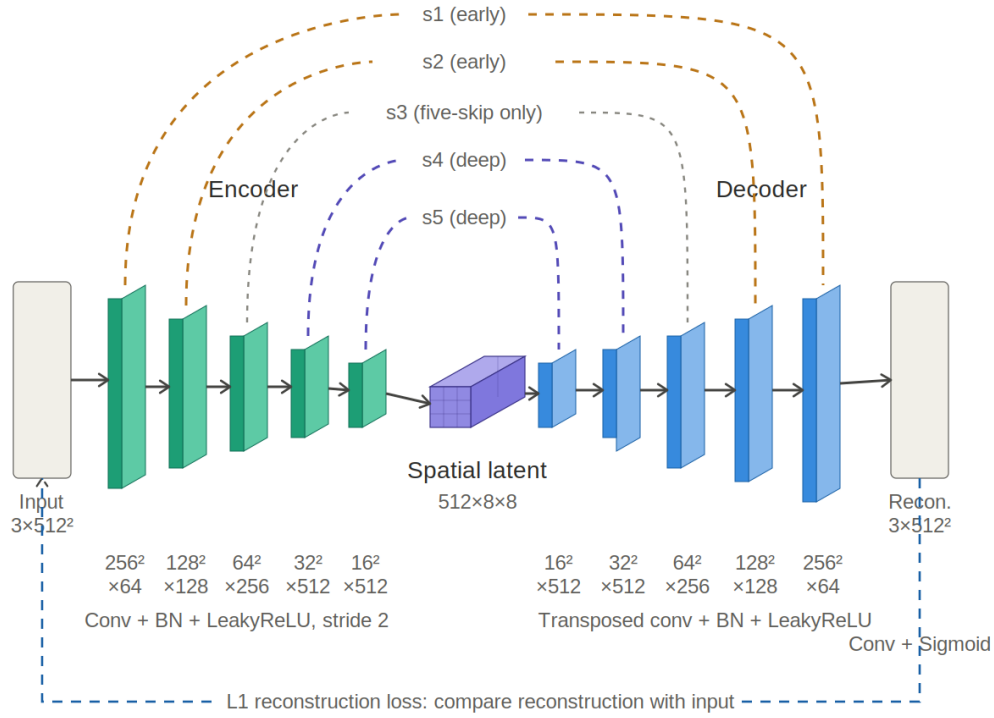
restrictive for a model without skip connections, since such a model has no alternative path between the encoder and decoder. Using a spatial latent keeps the comparison fair across models that differ in their skip-connection configuration, because no model is forced through a bottleneck that another model can bypass.

Within the CAE and diffusion mechanisms, several variants are trained that differ only in their skip connections. The motivation for this is specific to representation learning. Because evaluation uses the frozen encoder and a single global vector pooled from the spatial latent, the quality of that vector depends on how much information the encoder is required to carry through its bottleneck. Skip connections pass information from the encoder directly to the decoder, bypassing the bottleneck. If a large amount of information is bypassed in this way, the encoder is under less pressure to encode that information into the latent, and the pooled representation may therefore be weaker, even when reconstruction quality is good. The skip-connection variants are included to test this effect: they vary how much information can bypass the bottleneck, while keeping everything else fixed.

4.1 Convolutional Autoencoder (CAE)

The CAE consists of a convolutional encoder and a mirrored convolutional decoder. The encoder applies six stride-2 convolutional blocks, each a 3×3 convolution followed by batch normalisation and a LeakyReLU activation. The channel progression is $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 512 \rightarrow 512$, reducing a $3 \times 512 \times 512$ input image to a $512 \times 8 \times 8$ spatial latent. The decoder mirrors this structure with transposed convolutions that upsample the latent back to a $3 \times 512 \times 512$ reconstruction, ending in a sigmoid activation so that the output matches the $[0, 1]$ range of the input. The model is trained with an L1 reconstruction loss between the input and the reconstruction. For evaluation, the $512 \times 8 \times 8$ latent is mean-pooled over its spatial dimensions to give a 512-dimensional embedding.

Four CAE variants are trained, sharing the encoder, decoder, and training configuration above and differing only in their skip connections. Where a skip is present, the encoder feature map is concatenated with the decoder feature map at the matching resolution and combined by a 3×3 convolution. Skips at different depths bypass different kinds of information, so each configuration changes what the latent is no longer required to encode; the variants below are an open test of how this affects the frozen pooled representation. Figure 5 shows the architecture and the skip connection positions used by the four variants.



Skip connections

- Early skips (s1, s2) — high resolution, fine detail
- Deep skips (s4, s5) — low resolution, coarse layout
- s3 — active only in the five-skip variant
- Data flow (forward path)

Variants (which skips are active)

- No-skip — none; all data through the latent.
- Two early skips — s1, s2.
- Two deep skips — s4, s5.
- Five skips — s1, s2, s3, s4, s5.

Block height indicates spatial resolution; printed values are resolution $\times$ channels. A skip concatenates an encoder feature map with the decoder stage at the same resolution, then a 3×3 convolution merges them.

Figure 5. The convolutional autoencoder, showing the six-block encoder, the $512 \times 8 \times 8$ latent, the mirrored decoder, and the skip-connection positions used by the four variants.

4.1.1 No-Skip

No skip connections are used, so every part of the reconstruction must be produced from the $512 \times 8 \times 8$ latent alone. This variant is the reference point for the comparison: it places the full reconstruction burden on the latent, so the representation it learns reflects only what the bottleneck itself can retain. Any difference between this variant and the skip variants isolates the effect of allowing information to bypass the bottleneck.

In representation terms, this is the configuration under the most pressure to compress: because nothing can be passed around the latent, the encoder must store in it whatever is needed to reconstruct the image, including both coarse structure and fine detail. It therefore provides the cleanest test of how much useful structure the latent can hold on its own.

4.1.2 Two Early Skips

Skip connections are placed at the two highest-resolution encoder stages only. Early stages carry fine, local information — edges, texture, and small-scale detail — at high spatial resolution. Passing these directly to the decoder means the latent is no longer required to preserve that fine detail in order to reconstruct it, since the decoder can obtain it from the skip.

This variant therefore tests what happens to the representation when fine local detail is offloaded from the bottleneck. Such detail is often the least relevant to disease-level structure, so removing the need to encode it could either leave the representation unchanged, or free the latent to allocate its capacity to coarser, more diagnostic structure. It is included to probe that question rather than to assume an outcome.

4.1.3 Two deep Skips

Skip connections are placed at two deeper, lower-resolution encoder stages instead of the early ones. Deep stages carry coarse, abstract information — large-scale spatial layout rather than fine texture. Passing these around the latent means the latent is relieved of coarse structural information instead of fine detail.

This variant is the counterpart to the early-skip configuration: the two together test whether the depth at which information bypasses the bottleneck matters, by relieving the latent of opposite kinds of information while holding the number of skips fixed at two. Comparing them indicates whether it is fine detail or coarse layout whose removal more strongly affects the frozen representation.

4.1.4 All Five Skips

Skip connections are placed at all five encoder stages above the latent, so information at every resolution has a path to the decoder that does not pass through the bottleneck. This is the configuration in which the latent is under the least pressure, because almost any information the decoder needs can be supplied directly by a skip.

It represents the upper end of the range the variants span, from no bypass (no-skip) to bypass at every level (five skips). If allowing information to bypass the bottleneck weakens the pooled representation, the effect should be most visible here; if it does not, this variant shows that strong reconstruction through skips does not in itself degrade the representation.

4.2 Diffusion-Based Model

The diffusion model is a denoising network trained according to the denoising diffusion probabilistic model formulation. During training, a clean image is corrupted by adding Gaussian noise at a randomly sampled timestep using a fixed linear noise schedule, and the network is trained to predict the noise that was added. The training loss is the L1 difference between the predicted noise and the true noise. The timestep is provided to the network through a sinusoidal timestep embedding that is added as a channel-wise bias within each block.

The encoder (down path) follows the same six-block, stride-2 structure and the same channel progression as the CAE encoder, reducing the noisy input to a $512 \times 8 \times 8$ representation, followed by a middle block at the same resolution. This middle block is the latent used for representation evaluation. The decoder (up path) upsamples back to a $3 \times 512 \times 512$ output, which is the predicted noise rather than a pixel image, so no sigmoid activation is applied at the output.

Because the model operates on noisy inputs, embedding extraction is defined to be deterministic. For evaluation, each image is corrupted at a single fixed timestep ($t = 250$) using the same noise schedule, passed through the encoder and middle block, and the resulting $512 \times 8 \times 8$ latent is mean-pooled to a 512-dimensional embedding. The same fixed timestep is used for every diffusion variant so that their embeddings are produced under identical conditions.

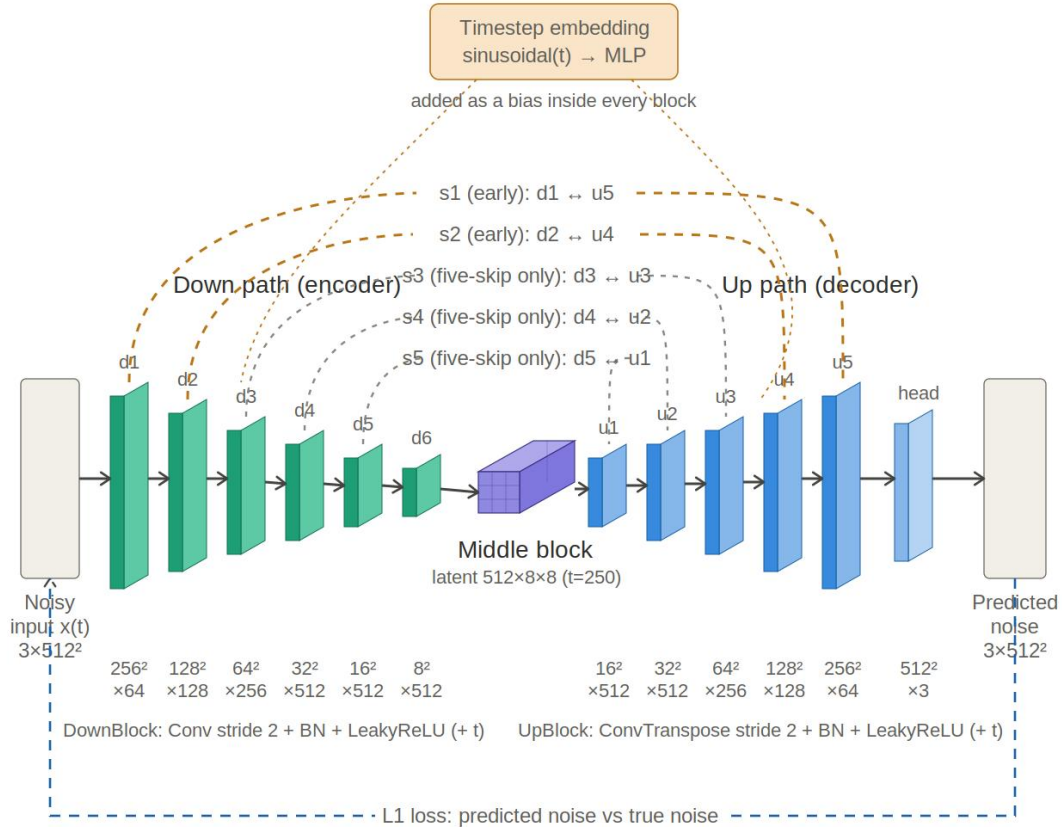
Three diffusion variants are trained, differing only in their skip connections between the down path and the up path, with the encoder, decoder, and training configuration otherwise fixed. Where a skip is present the down-path feature map is concatenated with the corresponding up-path feature map and combined by a convolution; where it is absent the up-path block upsamples without concatenation. As with the CAE, the placement of skips determines what the middle-block latent is no longer required to encode, and the variants test how this affects the frozen representation. Figure 6 shows the architecture and the skip connection positions used by the three variants.

4.2.1 No-Skip

No skip connections are used between the down path and the up path, so the predicted noise is produced from the middle-block latent alone. This matches the no-skip CAE and serves the same reference role for the same reasons.

4.2.2 Two early Skips

Skip connections are placed at the two highest-resolution stages only, matching the two-early-skip CAE.



Skip connections

- Early skips (s_1, s_2) — high resolution, fine detail
- s_3, s_4, s_5 — active only in the five-skip variant
- Timestep embedding fed into every block
- Data flow (forward path)

Variants (which skips are active)

- No-skip — no skips; all up blocks plain.
- Two early skips — s_1, s_2 (u_4, u_5 merge).
- Five skips — s_1 – s_5 (all up blocks merge).
- No deep-skip variant exists for the diffusion model.

Each skip joins a down block to the up block at the same resolution. Variants differ in block type, not only in arc presence: an up block with an active skip concatenates the skip and merges it with a convolution; a plain up block has no concat or merge. The down path, middle block, timestep embedding, and noise-prediction head are identical across all three variants.

Figure 6. The diffusion-based model, showing the down path, the middle-block latent, the up path predicting noise, the timestep embedding, and the skip-connection positions used by the three variants.

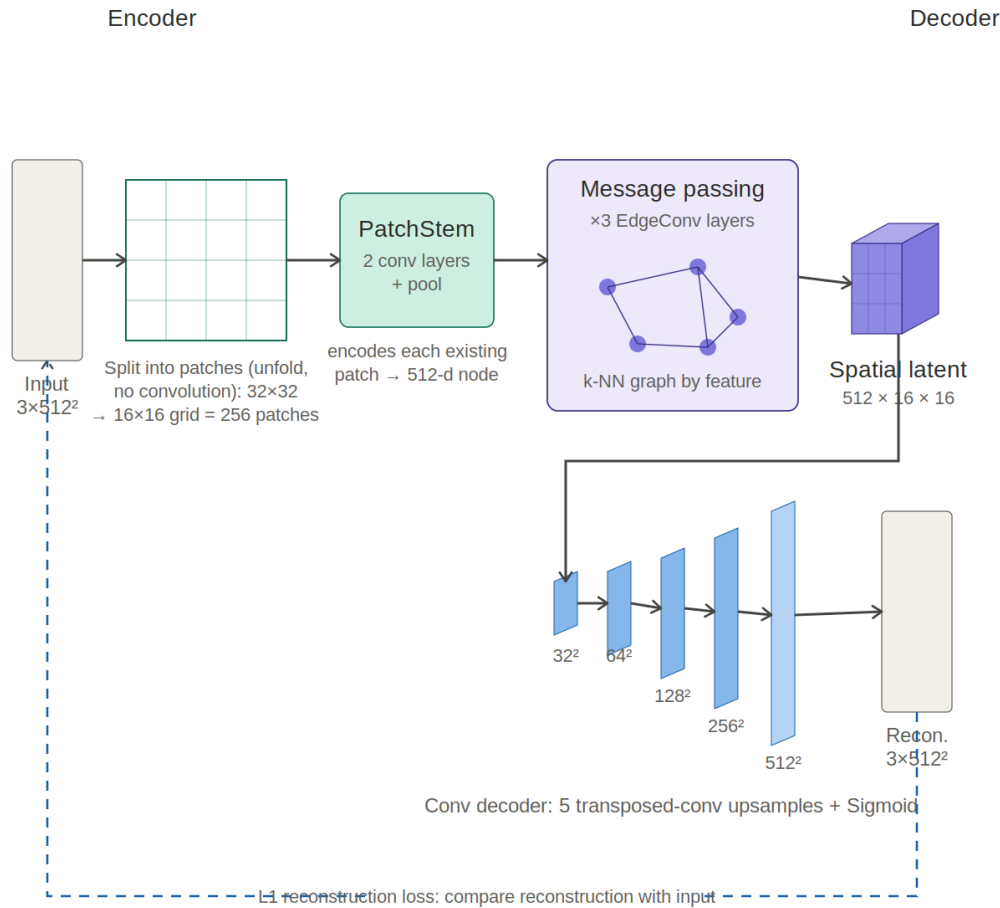
4.2.3 All Five skips

Skip connections are placed at all five stages above the latent, matching the five-skip CAE. With the no-skip variant, this bounds the diffusion comparison from no bypass to full bypass. The diffusion mechanism is not given a deep-skip variant, so the early-versus-deep contrast is examined only for the CAE.

4.3 Graph Based Autoencoder (GAE)

The GAE replaces the convolutional encoder with an encoder based on graph message passing, while keeping the same autoencoder structure and L1 reconstruction objective.

Each input image is divided into non-overlapping 32×32 patches, producing a 16×16 grid of 256 patches per image. Each patch is treated as a node in a graph. Figure 7 shows this architecture.



The image is first cut into fixed patches by an unfold operation (no convolution). The conv stem then encodes each patch into a 512-d node vector. A k-nearest-neighbour graph ($k = 10$) is built from node-feature similarity and rebuilt at each of the three EdgeConv layers. Updated nodes are placed back on the 16×16 grid to form the latent, which the decoder reconstructs.

Figure 7. The graph-based autoencoder, showing the patch split, the convolutional stem, the three dynamic k-nearest-neighbour message-passing layers, the $512 \times 16 \times 16$ latent, and the convolutional decoder.

A shallow convolutional stem converts each patch into a node feature vector. The stem applies two 3×3 convolutions ($3 \rightarrow 64 \rightarrow 512$), each with batch normalisation and a LeakyReLU activation, followed by adaptive average pooling that collapses the patch to a single 512-dimensional vector. The stem is deliberately shallow so that most of the representational work is performed by the graph message passing rather than by convolution within each patch.

The graph over the patch nodes is built dynamically from feature similarity rather than from spatial position. At each message-passing layer, the node features are L2-normalised and a k-nearest-neighbour graph is constructed in this normalised feature space ($k = 10$),

which connects each node to the ten nodes most similar to it; normalising before the nearest-neighbour search makes the similarity equivalent to cosine similarity. Because the graph is rebuilt from the current features at every layer, it is a dynamic graph that can change between layers and is not fixed to the spatial layout of the patches. This allows information to pass between regions of the image that are similar in appearance but not spatially adjacent, which a single convolution cannot do.

Three message-passing layers are applied. Each layer is an EdgeConv operation, in which a small multilayer perceptron processes, for each edge, the concatenation of a node’s features and the difference to each neighbour, and the results are aggregated over neighbours. Each layer is applied residually, so its output is added to its input. After message passing, the node features are arranged back onto the 16×16 grid to form a $512 \times 16 \times 16$ spatial latent. A convolutional decoder of five transposed-convolution upsampling blocks maps this latent back to a $3 \times 512 \times 512$ reconstruction, ending in a sigmoid activation. For evaluation, the $512 \times 16 \times 16$ latent is mean-pooled over its spatial dimensions to give a 512-dimensional embedding, consistent with the other models.

The selection of nearest neighbours that defines the graph is not differentiable: the node features are learned, but the choice of which nodes are connected is not trained directly. The GAE is not divided into variants; it is evaluated as a single configuration.

5. Evaluation

5.1 Frozen Representations

All models were evaluated as fixed feature extractors. After training, the weights of each model were frozen and a single embedding vector was produced for every image by taking the model’s spatial latent and averaging it over its spatial dimensions, giving a 512-dimensional vector per image. For the diffusion models the latent was taken from the middle block after adding noise at a single fixed timestep, so that the embeddings were deterministic. No part of the encoder was updated during evaluation. The methods below therefore measure the quality of the representations as learned during unsupervised training, without any further adaptation to the target task. The target task is the separation of age-related macular degeneration (ARMD), which was held out from training, from all other images.

All methods used the same locked test split, a balanced set of ARMD-positive and non-ARMD images selected with a fixed random seed. Using the same split for every model and every method makes the results directly comparable across models.

5.1.1 Linear Probe

The main measure of representation quality was a linear probe. A logistic-regression classifier was trained on the frozen embeddings to separate ARMD from non-ARMD, and

its performance was measured on the locked test set. A linear probe is used because it has limited capacity of its own: it can only draw a linear boundary in the embedding space, so its accuracy reflects how well the representation already separates the two classes rather than how powerful the classifier is. The area under the receiver operating characteristic curve (AUC) was used as the main metric, with classification accuracy reported alongside it. The classifier was weighted to account for the class imbalance in the training data.

To estimate how sensitive this measure is to the particular training sample, the probe was repeated five times, each time resampling the training set of 238 images with replacement (a bootstrap) to the same size, with a different random draw per repeat. The mean and standard deviation of the AUC across these five repeats are reported. This variation reflects sensitivity to the training sample; the test set was held fixed, so it does not capture variation due to the choice of test images.

5.1.2 Label-Purity

Some ARMD-positive images in the dataset carry additional disease labels, while others are labelled with ARMD alone. This raises the question of whether a model separates ARMD itself or instead responds to the other conditions that often accompany it. To examine this, the ARMD-positive test images were divided into two groups: those labelled with ARMD only, and those labelled with ARMD together with at least one other condition. The probe's performance was measured separately for each group against the same set of non-ARMD test images. Comparing the two indicates whether the separation is driven by ARMD-specific appearance or by co-occurring conditions. Because each group contains only a small number of images, these figures are treated as indicative rather than precise.

5.1.3 Raw-Pixel Baseline

To provide a reference point, the same linear probe was applied to a simple baseline representation that involved no learning. Each image was reduced to a small fixed size by downsampling and the resulting pixel values were used directly as the feature vector, with the same probe, the same locked test split, and the same bootstrap procedure as for the learned embeddings. This baseline sets a floor: it shows the level of ARMD separation obtainable from coarse colour and brightness information alone. A learned representation is only useful to the extent that it separates ARMD better than this floor, so the baseline gives a concrete threshold against which each model can be judged.

5.1.4 Cluster Separation

As an additional measure that does not depend on training a classifier, the silhouette score was computed over the embeddings using the binary ARMD-versus-rest labels. The silhouette score summarises how close each point is to others with the same label compared with points of the other label. It is reported as a secondary measure only: the dataset is imbalanced, which tends to keep the score low regardless of representation quality, so it is read as supporting context rather than as a primary result.

5.1.5 Embedding Visualization

To inspect the structure of the embeddings directly, the high-dimensional vectors were projected to two dimensions using t-SNE and UMAP and plotted. In each plot the points were coloured by three groups: ARMD-only, ARMD with at least one other condition, and not-ARMD. The large not-ARMD group was randomly subsampled so that it did not dominate the plot. These projections are qualitative: they are used to look for visible grouping of the ARMD points and to compare the overall structure produced by different models, not to provide a numerical result. Both methods were applied with fixed settings so that the plots are reproducible.

5.2 Fine-Tuning

In addition to the frozen evaluation, a fine-tuning experiment was carried out on the three top-performing models from the frozen evaluation: the diffusion model with five skip connections, the diffusion model with two early skip connections, and the graph-based autoencoder. The purpose of this experiment was to examine how the classification performance of each model changes as the number of labelled training examples increases. This question is sometimes called label efficiency: it asks how much labelled data is needed to obtain useful classification performance from a pretrained representation. The remainder of this section describes the protocol, the model configuration, the training procedure, and the evaluation.

5.2.1 Models Selected

Only the three models that exceeded the raw-pixel baseline in the frozen evaluation were taken forward. Fine-tuning the remaining five models was not pursued, because their frozen representations did not separate ARMD better than coarse downsampled pixels. For those models a fine-tuning result, regardless of its value, would not isolate the contribution of the learned representation from the contribution of the supervised fine-tuning step, so it would not address the research question.

5.2.2 Trainable Components

Only the last block of each encoder and a single linear classification head were updated during fine-tuning. The remainder of each encoder was kept frozen, with its pretrained parameters fixed. For the two diffusion models the last block was the middle block, which produces the latent used as the embedding in the frozen evaluation. For the graph-based autoencoder the last block was the final message-passing layer. The classification head was a single linear layer mapping the 512-dimensional mean-pooled latent to one output, identical in capacity to the linear probe used in the frozen evaluation.

This choice keeps the comparison between the frozen and fine-tuned conditions as direct as possible. The pretrained representation is retained almost in full, only the final stage is adapted, and the classifier on top has the same form as the frozen probe. The only variable between the two conditions is whether the last block is allowed to update during training.

5.2.3 Training Data and Label Budgets

The training data was drawn from the ARMD pool described in the data preprocessing section, with the locked test set held aside. From the remaining ARMD-positive and non-ARMD images, balanced training sets of four sizes were constructed: ten, twenty-five, fifty, and one hundred positives, each paired with the same number of non-ARMD images. These four sizes are referred to as label budgets. Random horizontal and vertical flips were applied to the training data; no other augmentation was used. The locked test set was not augmented.

For each budget, the training set was redrawn five times with different random seeds, giving five repeats per budget per model. The seeds were derived from the same fixed base seed used for the locked test split, so that the experiment is reproducible. Each repeat used a different random sample from the same underlying pool of remaining images, so the variation between repeats reflects sensitivity to which labelled examples were drawn rather than to the random initialisation of the model. At the smallest budget the training set was small enough that, after batch processing, the effective number of training images per epoch was slightly below the requested total; this is noted alongside the results.

5.2.4 Optimisation

Training used the Adam optimiser with a fixed learning rate of $1e-4$ for every model, every budget, and every repeat. This learning rate is lower than the rate used for unsupervised pretraining and reflects the intention of adapting the last block rather than retraining it from scratch. The loss was binary cross-entropy with logits, suited to the single-output linear head. Each run was trained for a fixed number of epochs and the model from the final epoch was used for evaluation. No early stopping or model selection during training was performed.

The reason for using a fixed schedule without early stopping is to avoid any use of the test set during training. With training pools as small as the smallest budgets here, it is not possible to set aside a meaningful validation split within the training set, and using the test set for stopping decisions would compromise the independence of the final evaluation. Using a fixed schedule for every run removes this risk and makes the protocol identical across models, budgets, and repeats, at the cost of accepting that the chosen epoch count may not be optimal for every configuration.

5.2.5 Evaluation

All fine-tuned models were evaluated on the same locked test set used in the frozen evaluation. The test set contains an equal number of ARMD-positive and non-ARMD images and was not used at any point during fine-tuning. For each run the trained model produced a probability for each test image, and from these probabilities two metrics were computed: the area under the receiver operating characteristic curve (AUC) and the classification accuracy at a threshold of 0.5. For each combination of model and budget, the mean and standard deviation of the AUC and accuracy were taken across the five

repeats. The use of the same test set as in the frozen evaluation allows the fine-tuned results to be compared directly with the frozen-probe results from the previous section.

6. RESULTS

6.1 Results of Frozen Representations

Each model was evaluated as a frozen feature extractor on the same locked test set. For every model the linear probe is reported as AUC and accuracy with bootstrap variation, alongside the label-purity breakdown, the raw-pixel floor, and the silhouette score. The raw-pixel floor of 0.810 is the same for every model because it depends only on the images and the fixed test split. The complete results for all eight models are given in the table below, and a comparison of the linear-probe AUC across models is shown in Figure 8.

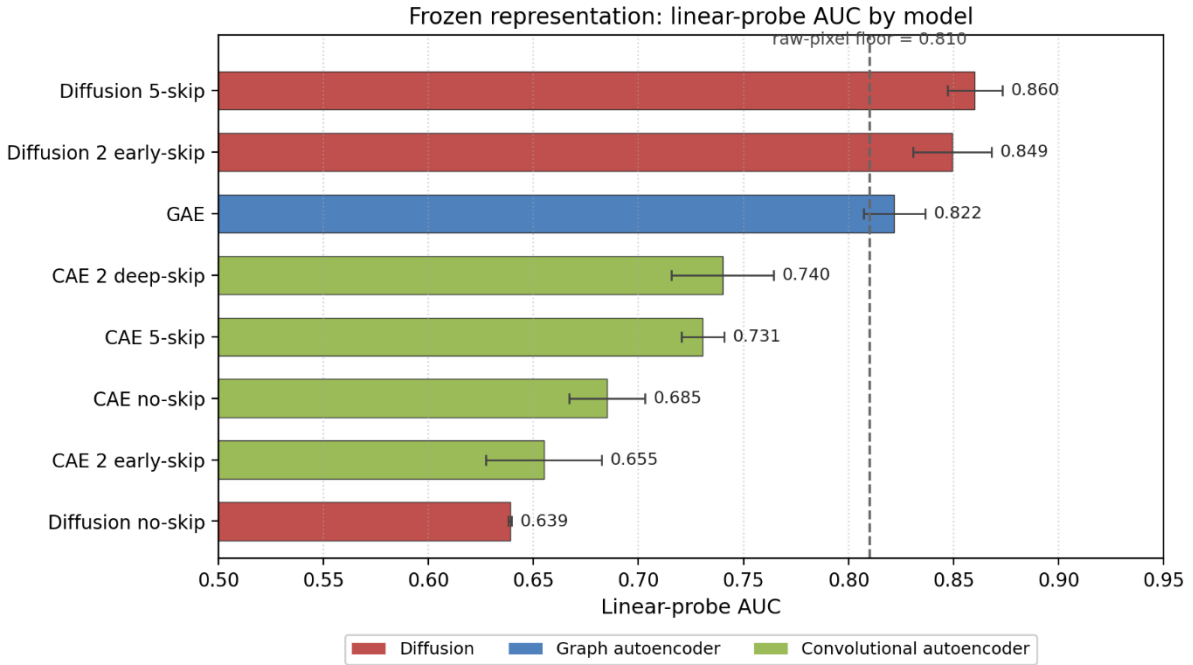


Figure 8. Linear-probe AUC for the eight frozen models on the locked test set, with the raw-pixel floor shown as a dashed line.

Model		Lin. AUC	Acc.		ARMD-only AUC	ARMD+other AUC	Floor AUC	Silhouette
Diffusion, skip	5-	0.860 $\pm$ 0.013	0.784		0.904	0.917	0.810	0.015
Diffusion, early-skip	2	0.849 $\pm$ 0.019	0.772		0.905	0.938	0.810	0.056
Graph autoencoder		0.822 $\pm$ 0.015	0.748		0.865	0.830	0.810	0.075
CAE, 2 deep-skip		0.740 $\pm$ 0.024	0.714		0.766	0.728	0.810	0.010
CAE, 5-skip		0.731 $\pm$ 0.010	0.656		0.699	0.792	0.810	0.165
CAE, no-skip		0.685 $\pm$ 0.018	0.654		0.731	0.657	0.810	0.043
CAE, 2 early-skip		0.655 $\pm$ 0.028	0.600		0.704	0.642	0.810	0.266
Diffusion, no-skip		0.639 $\pm$ 0.001	0.608		0.709	0.590	0.810	0.147

Table 1. Frozen-representation evaluation results for the eight models, ordered by linear-probe AUC. The raw-pixel floor AUC of 0.810 is the same for every model. Purity-group figures are based on sixteen ARMD-only and twelve ARMD-with-other test images and are indicative only.

Three models scored above the raw-pixel floor of 0.810: the diffusion model with five skips (0.860), the diffusion model with two early skips (0.849), and the graph-based autoencoder (0.822). The remaining five scored below it, comprising the four convolutional variants (0.655 to 0.740) and the diffusion model without skips (0.639). On the label-purity breakdown the three models above the floor scored between 0.83 and 0.94 on both the ARMD-only and ARMD-with-other groups. The diffusion model without skips recorded the lowest AUC (0.639) and the smallest bootstrap standard deviation of any model (0.001). The silhouette scores ranged from 0.010 to 0.266 and did not follow the linear-probe ordering.

The t-SNE and UMAP projections for the diffusion model without skips are shown in Figure 9. The points are arranged along a single continuous curved band rather than the more diffuse spread seen for the other models, and the ARMD and non-ARMD points are interleaved along this band.

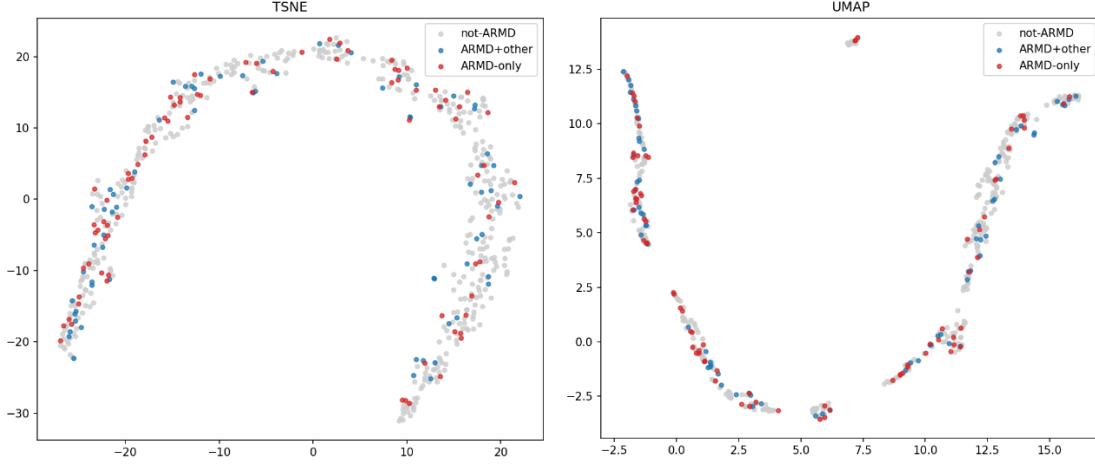


Figure 9. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the diffusion model without skip connections.

The t-SNE and UMAP projections for the remaining seven models, and the training and validation reconstruction-loss curves for all models, are provided in Appendix A.

6.2 Fine-Tuning

This section reports the fine-tuning results for the three models taken forward from the frozen evaluation: the diffusion model with five skip connections, the diffusion model with two early skip connections, and the graph-based autoencoder. Each model was fine-tuned at four label budgets of ten, twenty-five, fifty, and one hundred ARMD-positive examples, with five repeats per budget, and evaluated on the same locked test set used for the frozen evaluation. For each budget the mean and standard deviation of the AUC and the accuracy across the five repeats are reported. At the smallest budget of ten positives, the training set after removal of an incomplete final batch contained sixteen images per repeat rather than twenty, of which between six and ten were ARMD positive; the figures at this budget reflect that reduced training size.

Model	B10 AUC	B10 Acc	B25 AUC	B25 Acc	B50 AUC	B50 Acc	B100 AUC	B100 Acc	Frozen AUC
Diffusion, 5-skip	0.659 ± 0.043	0.610 ± 0.038	0.674 ± 0.019	0.620 ± 0.028	0.712 ± 0.023	0.648 ± 0.017	0.741 ± 0.026	0.668 ± 0.044	0.860

Model	B10 AUC	B10 Acc	B25 AUC	B25 Acc	B50 AUC	B50 Acc	B100 AUC	B100 Acc	Frozen AUC
Diffusion, 2 early-skip	0.683 $\pm$ 0.024	0.608 $\pm$ 0.012	0.689 $\pm$ 0.019	0.630 $\pm$ 0.025	0.713 $\pm$ 0.022	0.650 $\pm$ 0.018	0.736 $\pm$ 0.022	0.658 $\pm$ 0.025	0.849
Graph autoencoder	0.708 $\pm$ 0.043	0.662 $\pm$ 0.061	0.742 $\pm$ 0.027	0.688 $\pm$ 0.025	0.825 $\pm$ 0.015	0.770 $\pm$ 0.021	0.859 $\pm$ 0.006	0.802 $\pm$ 0.018	0.822

Table 2. Fine-tuned AUC and accuracy (mean $\pm$ SD over five repeats) for the three models at each label budget, with the frozen-probe AUC included for comparison. Budgets are the number of ARMD-positive examples in the balanced training set.

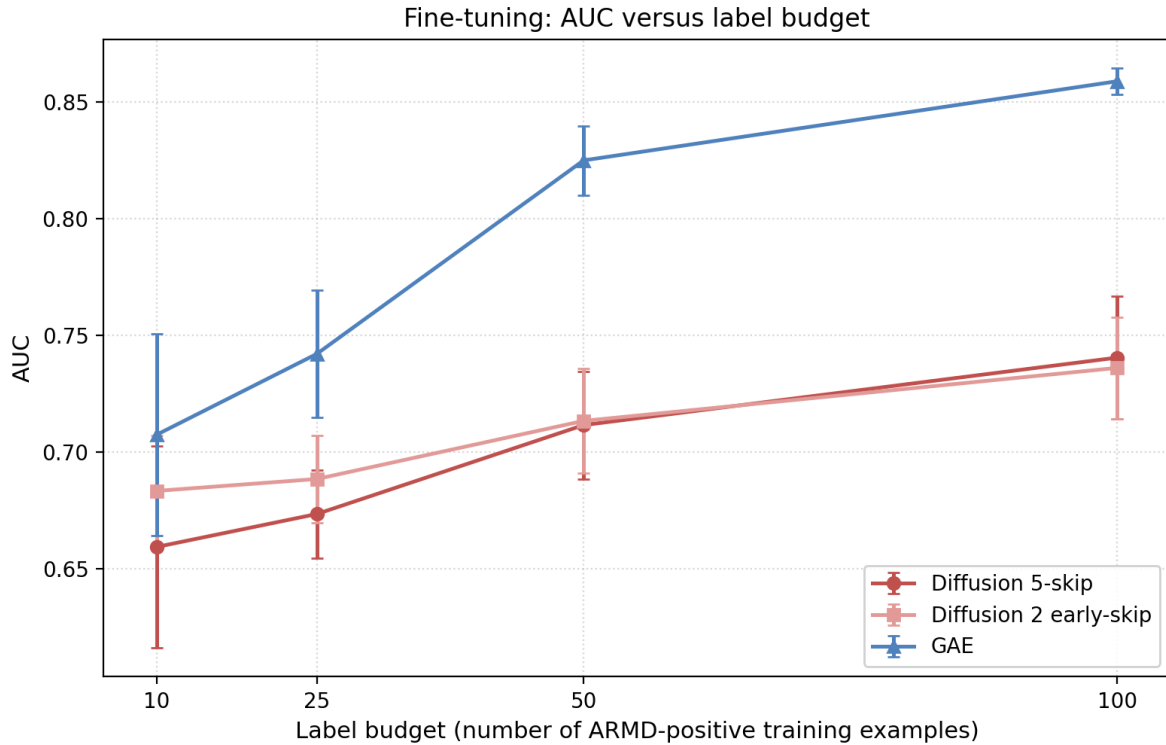


Figure 10. Fine-tuned AUC against label budget for the three models, with error bars showing the standard deviation across five repeats.

The fine-tuned AUC increased with the label budget for all three models, as shown in Figure 10. The graph-based autoencoder rose from 0.708 at ten positives to 0.859 at one hundred positives, exceeding its frozen-probe value of 0.822 at the largest budget. Both diffusion models remained below their frozen-probe values at every budget: the five-skip model reached 0.741 against a frozen value of 0.860, and the two-early-skip model reached 0.736 against a frozen value of 0.849. For all three models the standard deviation across repeats was largest at the smallest budget.

7. Analysis

This section interprets the results reported in the previous section. It considers the overall pattern across the eight models, the effect of the encoder mechanism, the effect of skip connections within each mechanism, the label-purity results, and the secondary measures. Each result is compared against the raw-pixel baseline and assessed relative to the reported variation and the size of the test set. The conclusions apply to the conditions used in this study: one held-out disease, one combination of datasets, and a test set of one hundred images.

7.1 Comparison Against the Raw-Pixel Baseline

The comparison against the raw-pixel baseline is the most informative result. A linear probe applied to coarse downsampled pixels, with no learning, reached an AUC of 0.810 on the locked test set. This value is high, which is consistent with ARMD being associated with large-scale changes in the retina, such as differences in colour and brightness near the macula, that remain present after downsampling. Each learned representation is therefore assessed against this value rather than against chance.

Relative to this baseline, the eight models divide into two groups. Three models scored above it: the diffusion model with five skips (0.860), the diffusion model with two early skips (0.849), and the graph-based autoencoder (0.822). The other five scored below it: the four convolutional variants (0.655 to 0.740) and the diffusion model without skips (0.639). This division is the main result of the study. It indicates that the choice of encoder mechanism, rather than the use of unsupervised reconstruction by itself, determines whether a representation separates ARMD better than raw pixels.

The four convolutional autoencoders did not separate ARMD better than downsampled pixels. This is a result about the convolutional mechanism under the conditions tested, not a fault in the procedure. It is not clear why this happened. The reconstruction objective trains the encoder to keep the information needed to rebuild the image, and this should include the same large-scale colour and brightness that simple averaging keeps. On that basis the mean-pooled latent might be expected to separate ARMD at least as well as the downsampled pixels. This was not the case: across all four variants the latent scored below the raw-pixel baseline. One possibility is that the information is present in the latent but not in a form that a linear probe can use, and another is that mean pooling discards part of it, but these experiments do not test either idea. The cause is therefore left as an open question.

7.2 Effect of the Encoder Mechanism

Comparing the strongest variant of each mechanism gives a consistent ordering. The two diffusion variants with skips and the graph autoencoder hold the three highest positions, separated from the convolutional variants by a margin larger than the reported variation: the strongest convolutional model reached 0.740, while the lowest of the three leading models reached 0.822. The diffusion mechanism produced the highest scores, the graph

mechanism was lower but still above the baseline, and the convolutional mechanism produced the lowest scores.

One possible reason for the higher diffusion scores is the training objective. A denoising model predicts the noise added to an image across a range of corruption levels, which requires it to represent how retinal structure is arranged and how it should appear when partly obscured. This task is more demanding than direct reconstruction and may lead the encoder to represent structure in a form that is more useful for separating an unseen disease. The graph mechanism connects image regions by feature similarity rather than only by spatial position, and its scores suggest that relating non-adjacent regions is useful for this task. These explanations are consistent with the results but are not directly tested by them. The study measures outcomes rather than the internal cause of those outcomes, so the explanations are interpretations rather than established findings.

7.3 Effect of Skip Connections

Skip connections had different effects in the two mechanisms where they were varied. For the diffusion model, adding skips increased the AUC from 0.639 without skips to 0.860 with five skips, a difference of 0.221. For the convolutional model, the same comparison increased the AUC from 0.685 to 0.731, a difference of 0.045. The effect of skip connections was therefore large for the diffusion model and small for the convolutional model.

This pattern is not consistent with a simple form of the bottleneck argument, which holds that allowing information to bypass the latent should weaken the representation because the encoder is under less pressure to compress information into it. Under that argument the models with the most skips would produce the weakest representations. For the diffusion model the opposite occurred, as the five-skip variant produced the highest score. A possible explanation is that the skips help the diffusion model train and predict noise more effectively, which produces a more informative middle-block latent, and that this effect is larger than any reduction in compression pressure. The convolutional model gained little from skips and so does not clearly support or contradict the argument. The result indicates that the effect of skip connections depends on the mechanism to which they are added and cannot be assumed in advance.

Within the convolutional family the four variants scored as follows: deep-skip 0.740, five-skip 0.731, no-skip 0.685, and early-skip 0.655. Several of these differences are within or close to the reported variation, so this ordering is not a reliable ranking. The available evidence supports only the statement that the four convolutional variants performed at a similar level and that all four scored below the raw-pixel baseline.

7.4 The Two Highest-Scoring Diffusion Models

The diffusion five-skip and diffusion two-early-skip models scored 0.860 and 0.849. Their reported standard deviations overlap, so the difference between them is small relative to the variation in the measurement, and they are treated as equal rather than ranked. The two-early-skip model uses fewer skip connections than the five-skip model but reached a similar score, which indicates that the two highest-resolution skips account for most of the effect and that the remaining three skips add little. On the available evidence either model would be a reasonable choice if a single configuration were selected for further work.

7.5 Label Purity

The label-purity breakdown was used to test whether each model separates ARMD itself or instead responds to other conditions that frequently occur alongside ARMD in the data. For the three models above the baseline, the AUC was high on both the ARMD-only group and the ARMD-with-other-condition group, with all three scoring between 0.83 and 0.94 on both groups. This indicates that their separation is not only a result of co-occurring diseases, because they separate ARMD-only images from non-ARMD images at a similar level to co-labelled ARMD images. This supports the interpretation that the three leading models represent ARMD-relevant structure.

For the models below the baseline the purity figures were lower and less consistent between the two groups, which matches their weaker overall separation. Each group contains only sixteen and twelve images, so the individual figures carry wide uncertainty and are used only to support the broader interpretation rather than as precise quantities.

7.6 Secondary Measures and an Anomaly

The silhouette scores were low for all models, from near zero to about 0.27, and did not follow the linear-probe ordering. For example, one of the weakest models by AUC produced one of the higher silhouette scores. This is expected, because the silhouette score is computed over an imbalanced binary split and tends to stay low regardless of representation quality. It is therefore used as supporting information rather than as evidence for or against any model.

One result requires specific comment. The diffusion model without skips produced the lowest AUC (0.639) and also an unusually small bootstrap standard deviation (0.001), smaller than that of any other model. A near-zero spread across resampled training sets indicates that the probe reached almost the same solution on every resample, which is consistent with embeddings that contain little information the probe can use, so that the result is largely insensitive to the training sample. Together with its low score and its position below the baseline, this configuration is best treated as weak or possibly under-trained, and its value should be interpreted with that limitation in mind.

The embeddings for this model also projected differently from the others, forming a single curved band in both the t-SNE and UMAP plots rather than a diffuse spread. One reading

is that the embeddings occupy a low-dimensional region of the latent space, which would be consistent with their small bootstrap spread and weak separation. Whether the latent is in fact low-dimensional was not tested, as the intrinsic dimensionality of the embeddings was not measured, so this is noted as an observation rather than an established cause.

7.7 Summary of the Frozen Representation

The results support three statements. First, the encoder mechanism affects representation quality: the diffusion and graph mechanisms produced representations that separate an unseen disease better than raw pixels, while the convolutional mechanism did not. Second, the effect of skip connections depends on the mechanism, with a large effect for the diffusion model and a small effect for the convolutional model, which indicates that this architectural choice does not transfer directly between mechanisms. Third, the separation produced by the three leading models reflects ARMD-relevant structure rather than co-occurring conditions, to the extent that the small purity groups allow this to be assessed. These statements apply to the conditions tested in this study and would require confirmation on other diseases, other datasets, and larger test sets before being treated as general. The complete set of figures is given in Table 1 in Section 6.1.

7.8 Overall Pattern

The three models did not respond to fine-tuning in the same way. The graph-based autoencoder improved over its frozen probe at the larger budgets and matched it at the smaller ones. The two diffusion models did not match their frozen probe at any budget; their fine-tuned AUCs were below the frozen value across all four budgets by a margin that exceeds the reported variation. The fine-tuning experiment therefore produced two distinct outcomes within a single experimental setup, depending on the model.

7.9 Graph Autoencoder

The graph-based autoencoder showed a clean upward trend in AUC as the label budget increased, from 0.708 at ten positives to 0.859 at one hundred positives. The standard deviation across repeats decreased as the budget grew, from 0.043 at the smallest budget to 0.006 at the largest, which is consistent with smaller training sets producing more variable results. At the largest budget the fine-tuned AUC of 0.859 is above the frozen-probe value of 0.822 by 0.037, a difference larger than the reported standard deviation at that budget.

This outcome indicates that updating the last message-passing layer together with the classification head allowed the model to use additional supervised information in a way that improved separation of ARMD. The pattern is consistent with a representation that was already useful in its frozen form and could be adapted further with a small number of labels. No catastrophic drop occurred at any budget, and the curve was monotonic

across all four budgets, which suggests that the training procedure was stable for this model under the conditions used.

7.10 Diffusion Models

The two diffusion models behaved similarly to each other and differently from the graph-based autoencoder. For the five-skip variant the fine-tuned AUC reached 0.741 at the largest budget, compared with a frozen-probe value of 0.860, a difference of 0.119. For the two-early-skip variant the fine-tuned AUC reached 0.736, compared with a frozen-probe value of 0.849, a difference of 0.113. Both differences are large relative to the reported variation. The AUC did increase with budget for both diffusion models, but the trajectory was lower than for the graph-based autoencoder and did not reach the frozen-probe value at any tested budget.

A possible explanation is that the diffusion encoder was pretrained to predict noise across a wide range of corruption levels, while the frozen evaluation and the fine-tuning both extract embeddings at a single fixed noise level. Updating the bottleneck and the classification head for a single-noise-level classification task may move the encoder away from the representation that the broader pretraining objective produced, even though that broader representation is what made the frozen probe perform well. This is offered as a possible reading and is not directly tested by the experiment. What the experiment shows is that, under the configuration used here, supervised fine-tuning did not produce a model better than the frozen probe for the diffusion variants.

7.11 Effect of Label Budget

For every model, the AUC at the smallest budget of ten positives was noticeably lower than at the larger budgets, and the standard deviation at this budget was the highest. This is consistent with the very small effective training size at this level: after the data-loading step that discards incomplete final batches, the smallest budget produced a training set of sixteen images per repeat rather than twenty, of which between six and ten were ARMD positive. Results at this budget are reported but should be read with this limitation in mind. From twenty-five positives upward the standard deviation decreased and the AUC values became more stable across repeats.

The pattern at the smallest budgets fits the idea of overtraining when very few labelled examples are available. At the ten positive budget the graph-based autoencoder reaches a higher AUC (0.708) than either diffusion model (0.659 and 0.683), which suggests it copes better with very small fine-tuning sets. When more of the blocks in the diffusion model were unfrozen instead of only the middle block, the test AUC went down rather than up at all four budgets. Letting more of the encoder change made the result worse, which is what happens when a model fits a small training set too closely and does not generalise. This is also why the main experiment trained only the last block.

The shape of the budget curves also differs between models. The graph-based autoencoder showed a substantial increase between fifty and one hundred positives,

suggesting that further labels would likely continue to help. Both diffusion models showed a smaller change between the two largest budgets, suggesting that under this protocol they were closer to a ceiling that lay below the frozen-probe value. Without more labels available it is not possible to say whether the diffusion curves would eventually reach the frozen-probe level given additional data.

7.12 Summary of Fine-Tuning

The fine-tuning experiment produced one model in which the procedure was useful and two in which it was not. For the graph-based autoencoder, partial fine-tuning of the last block together with the classification head improved on the frozen probe and the improvement grew with the label budget. For both diffusion models the same procedure produced AUCs below the frozen probe at every budget. Within the limits of this study, the frozen probe remains the best way to use the diffusion representations for ARMD separation, while the graph-based autoencoder benefits from supervised adaptation when labelled examples are available. The complete set of fine-tuning results is given in Table 2 in Section 6.2, alongside the corresponding frozen-probe AUCs for direct comparison.

8. CONCLUSION

This study compared three encoder mechanisms — convolutional, diffusion-based, and graph-based — for unsupervised representation learning on retinal fundus images, evaluated by how well their frozen representations separated an unseen disease, age-related macular degeneration, that was held out from training. The comparison was carried out under deliberately limited supervision, using a small fixed test set and a label-efficiency experiment.

The frozen evaluation produced a clear division. Measured against a raw-pixel baseline of 0.810 AUC, only the diffusion models with skip connections and the graph-based autoencoder separated the held-out disease better than coarse downsampled pixels; all four convolutional variants and the diffusion model without skip connections did not. Within this division, the diffusion mechanism produced the highest scores, the graph mechanism followed, and the convolutional mechanism did not exceed the baseline at any setting. Skip connections had a large effect on the diffusion model and only a small effect on the convolutional model, which indicates that the value of this architectural choice depends on the mechanism it is applied to rather than being general.

The fine-tuning experiment produced a different ordering from the frozen evaluation, and this difference is the central observation of the study. The graph-based autoencoder improved as more labelled examples were supplied and exceeded its own frozen result at the largest budget, whereas both diffusion models remained below their frozen results at every budget tested. The model with the strongest frozen representation was therefore

not the model that benefited most from supervised adaptation. This separates two properties that are easy to conflate: the quality of a representation when frozen, and its capacity to be improved by fine-tuning. A model can rank highly on one and poorly on the other. For the diffusion models in this study, the frozen representation remained the most effective way to use them; for the graph-based autoencoder, supervised adaptation was worthwhile when labelled data was available.

The high raw-pixel baseline is a feature of using ARMD as the held-out disease. ARMD produces large-scale colour and brightness changes that survive heavy downsampling, so simple averaging already separates it fairly well. This can be read two ways. As a limitation, it means the test disease is one where a trivial baseline is strong, so the headroom above the baseline is narrow and the comparison between models is made over a smaller margin. As a strength, it means the models that do exceed the baseline are separating ARMD on more than coarse colour and brightness, and are therefore learning something beyond what averaging captures. A disease that a trivial baseline did not already separate would give a wider margin but would not, on its own, show that the leading models learn anything more.

These findings hold only for the conditions tested. The study used a single combination of datasets, a single held-out disease, and a test set of one hundred images, with the label-purity groups smaller still; the smallest fine-tuning budget was further reduced by batch handling. The results should therefore be read as evidence from one controlled comparison rather than as general claims about these mechanisms. They do not establish clinical usefulness, which was not tested, and no claim about dementia is made. Within these limits, the study indicates that the encoder mechanism, and not merely the use of unsupervised reconstruction, determines whether a representation transfers to an unseen disease, and that frozen-representation quality and fine-tuning behaviour are distinct properties that should be assessed separately.

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Appendix A

A1. Embedding Projections

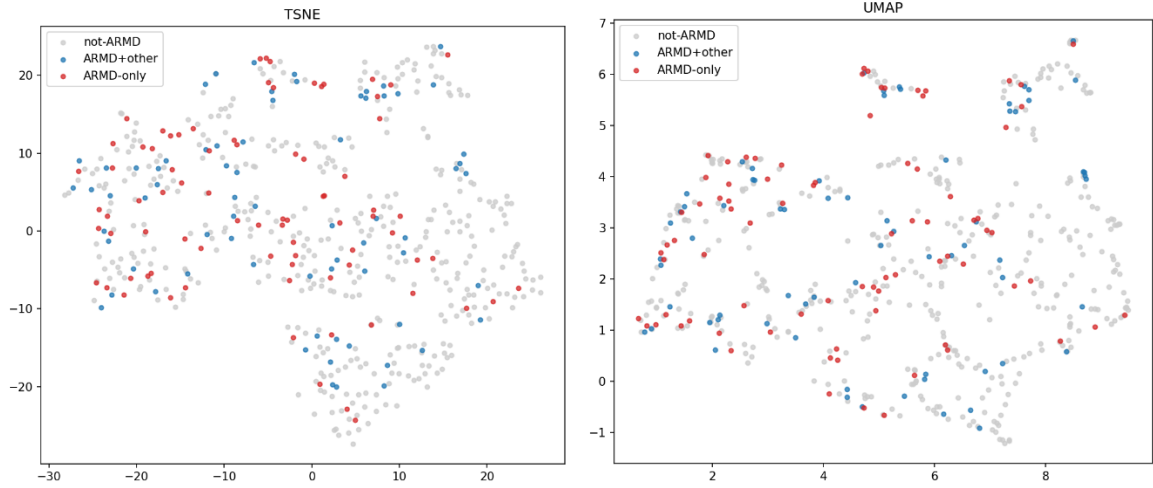


Figure A1. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the CAE No-Skip variant.

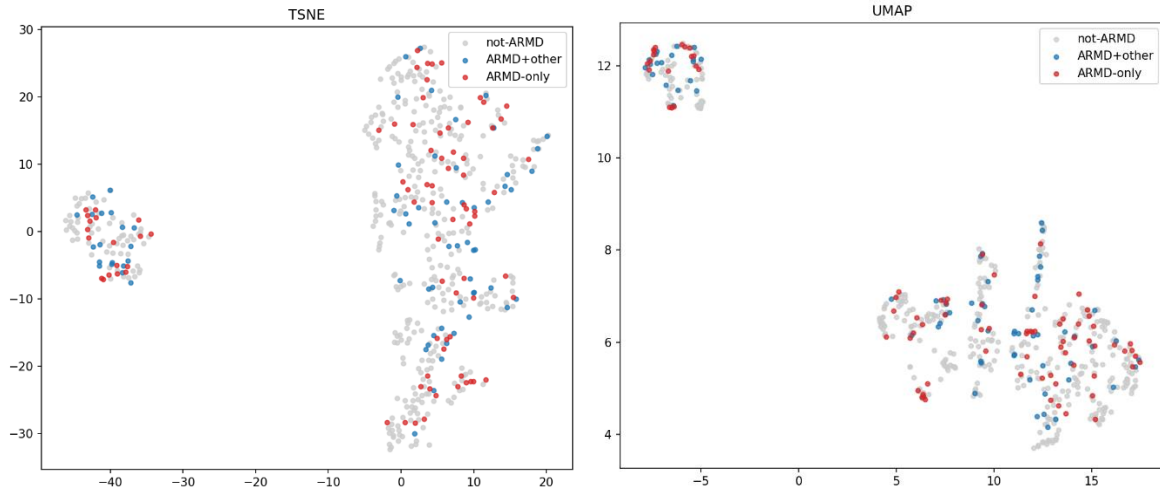


Figure A2. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the CAE All 5 Skip variant.

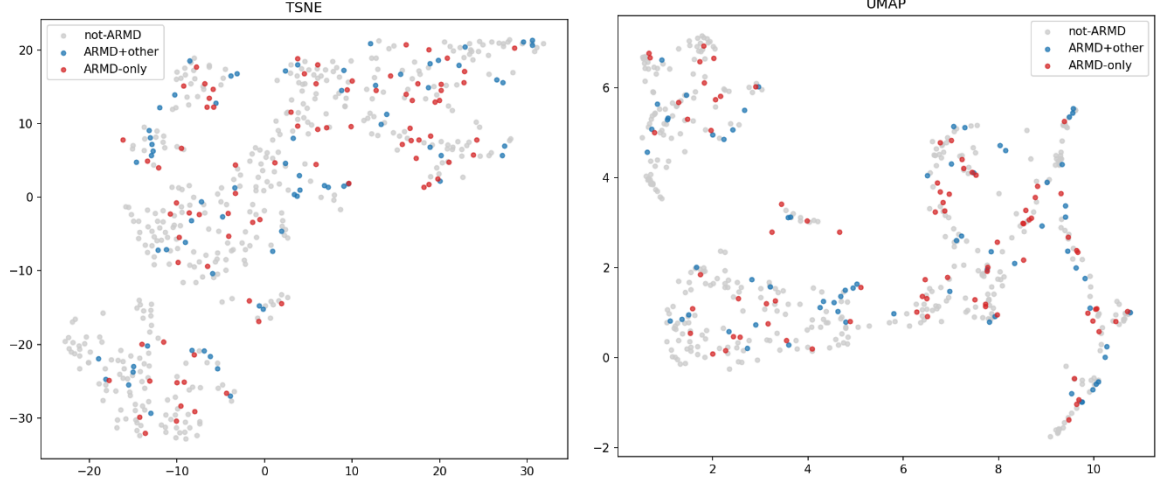


Figure A3. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the CAE Early Skip variant.

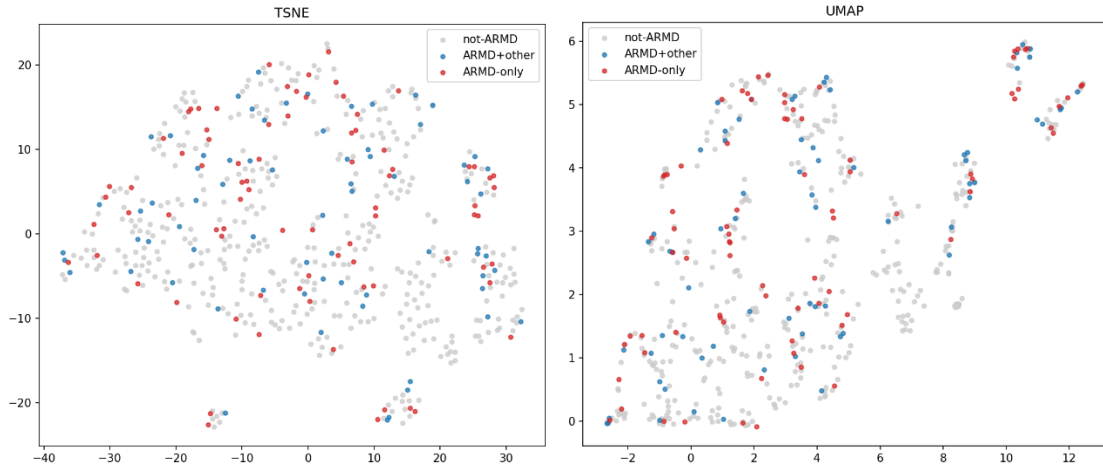


Figure A4. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the CAE Deep Skip variant.

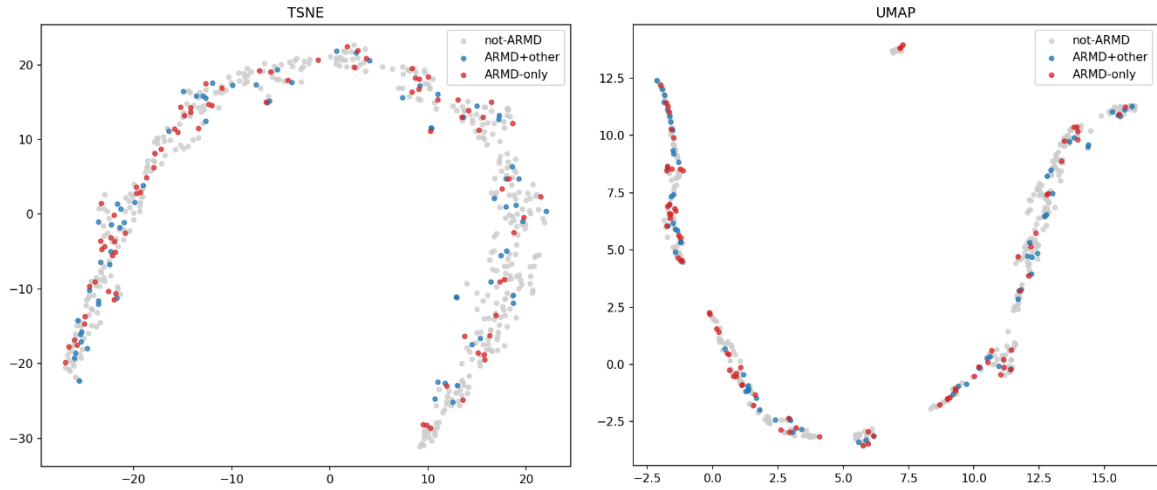


Figure A5. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the Diffusion No-Skip Variant.

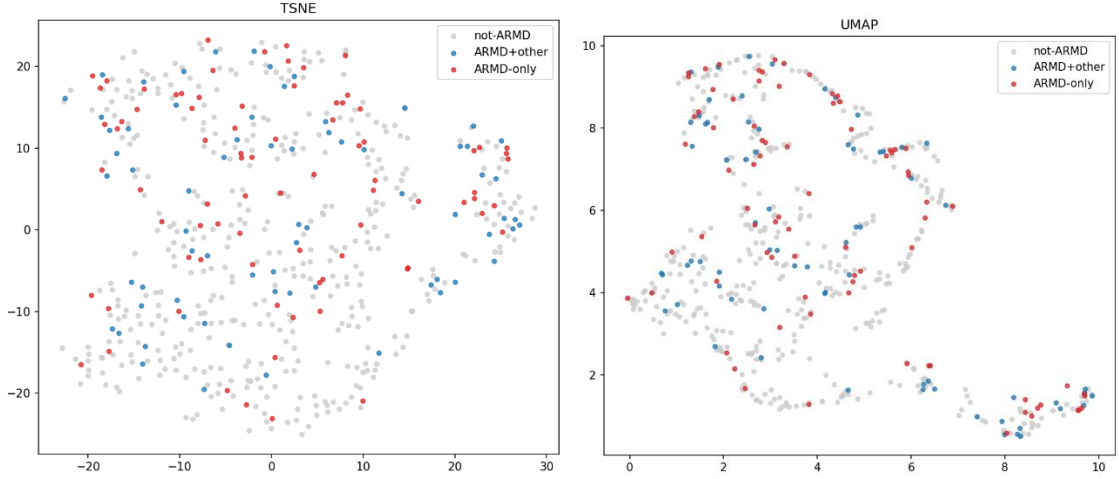


Figure A6. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the Diffusion Early Skip Variant.

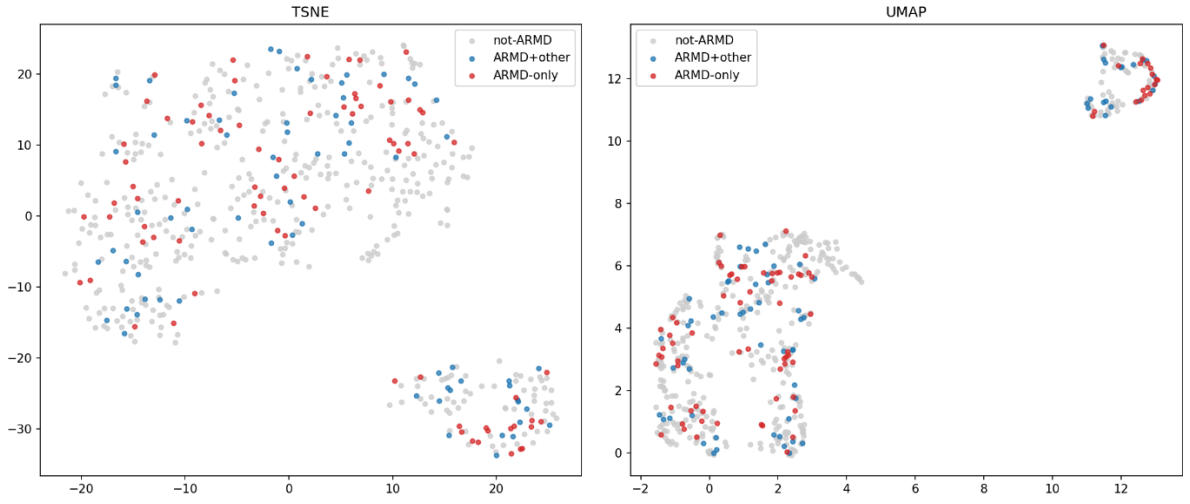


Figure A7. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the Diffusion All 5 Skip Variant.

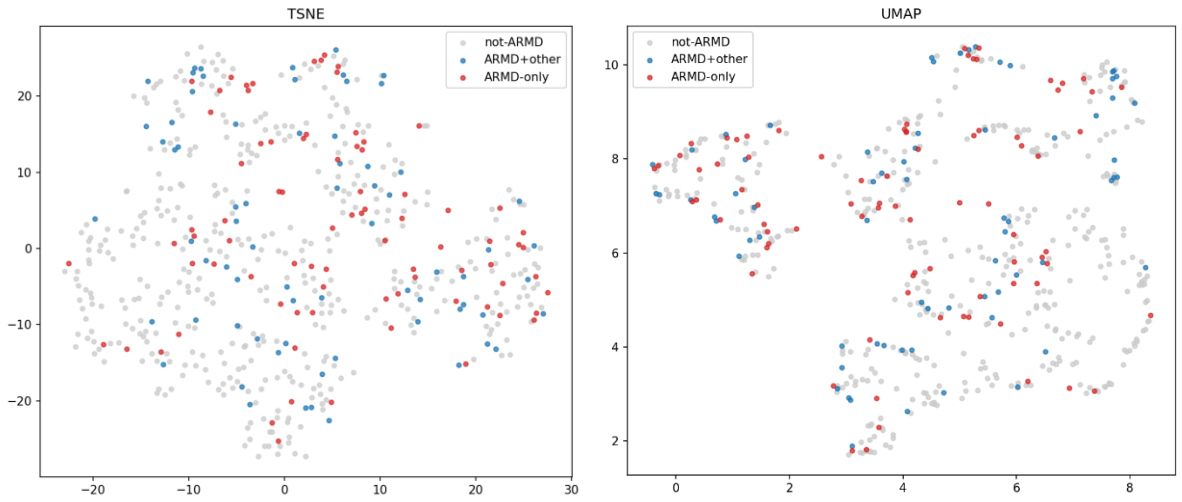


Figure A8. *t*-SNE (left) and UMAP (right) projections of the frozen embeddings for the GAE.

A2. Learning Curves

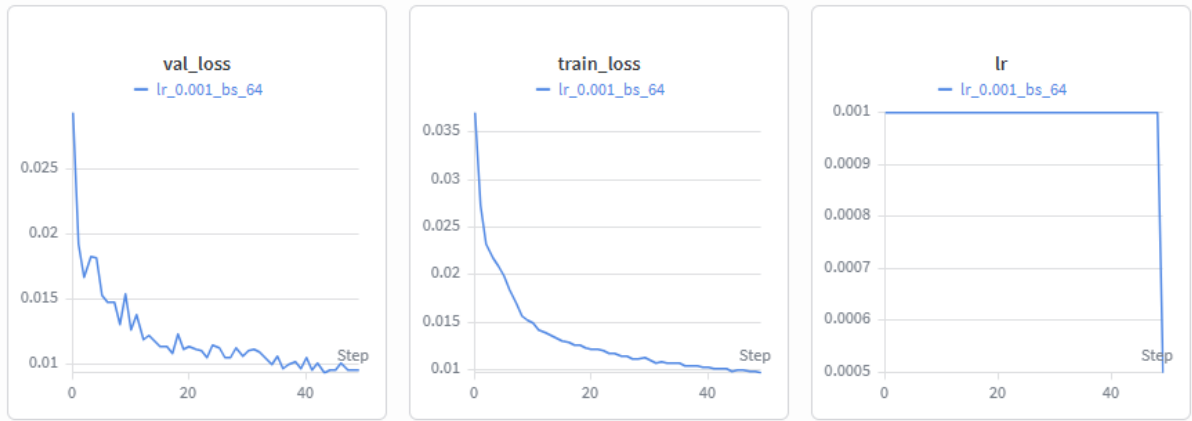


Figure A9. Training and validation reconstruction loss against epoch for the CAE No-Skip. The right panel shows the learning rate over training.

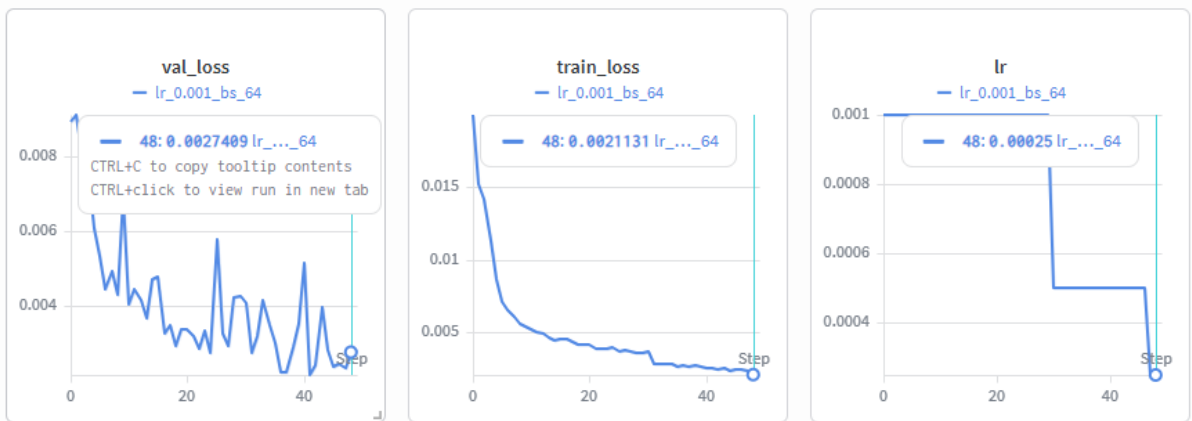


Figure A10. Training and validation reconstruction loss against epoch for the CAE All 5 Skips. The right panel shows the learning rate over training.

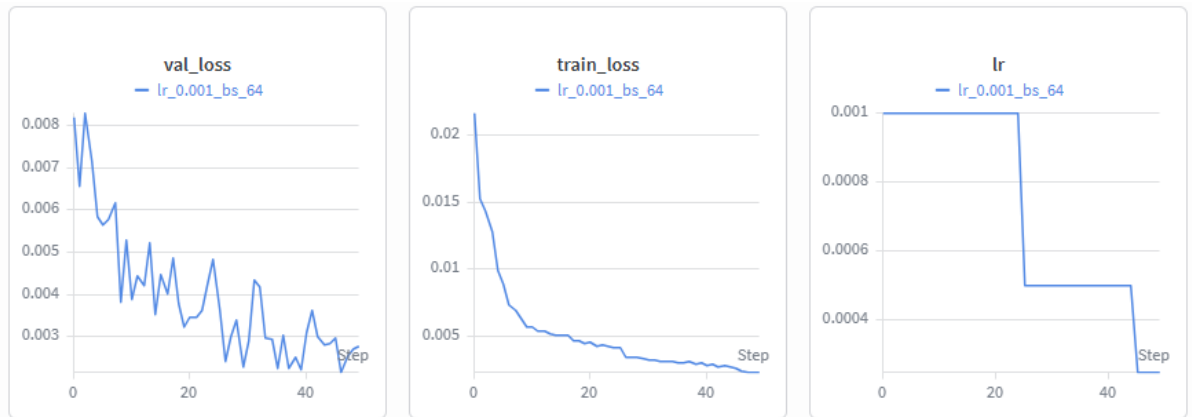


Figure A11. Training and validation reconstruction loss against epoch for the CAE Early Skip. The right panel shows the learning rate over training.

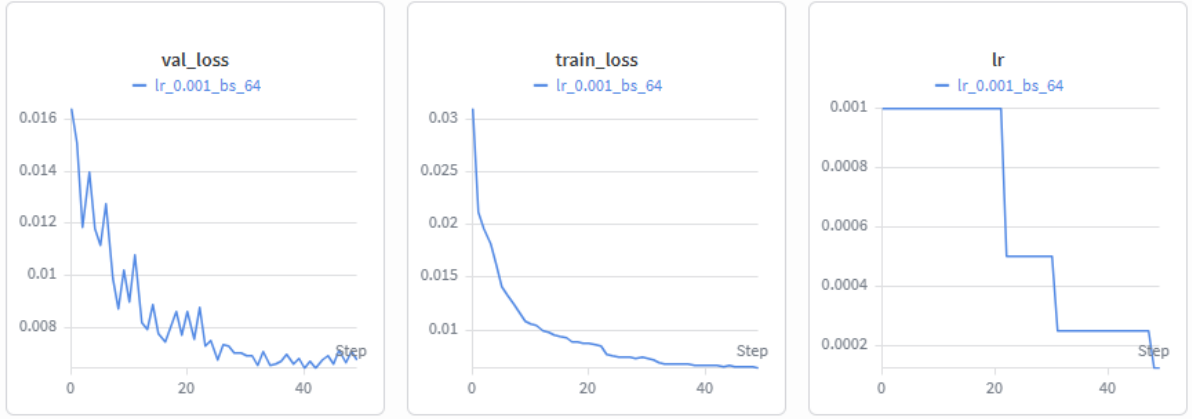


Figure A12. Training and validation reconstruction loss against epoch for the CAE Deep Skip. The right panel shows the learning rate over training.



Figure A13. Training and validation reconstruction loss against epoch for the Diffusion No Skip. The right panel shows the learning rate over training.

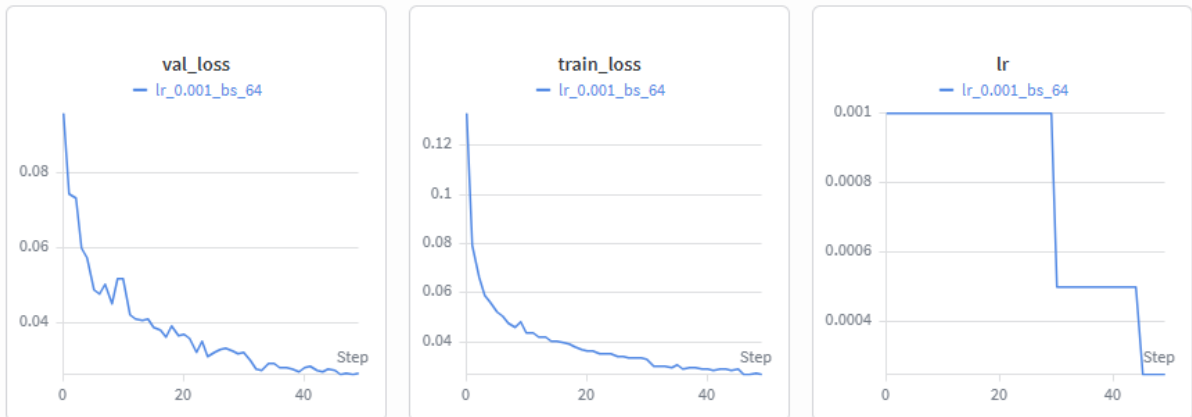


Figure A14. Training and validation reconstruction loss against epoch for the Diffusion Early Skip. The right panel shows the learning rate over training.

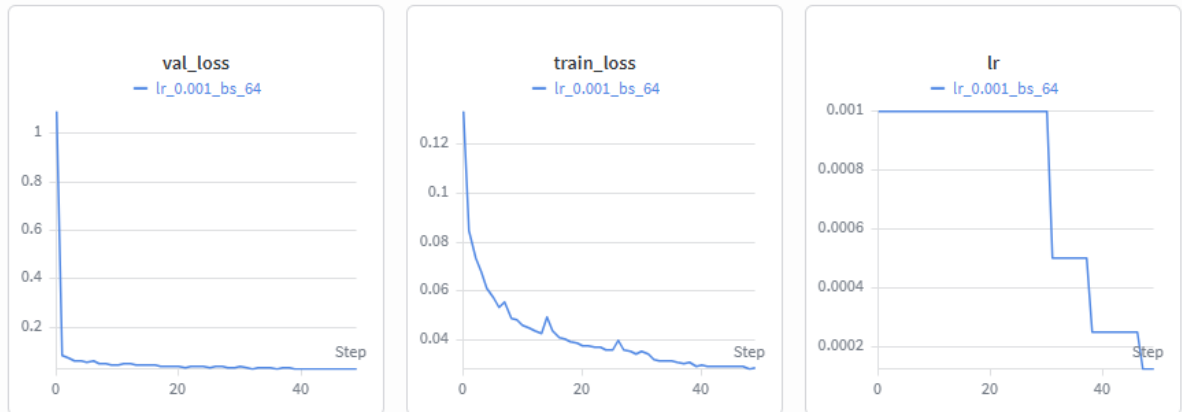


Figure A15. Training and validation reconstruction loss against epoch for the Diffusion All 5 skip. The right panel shows the learning rate over training.

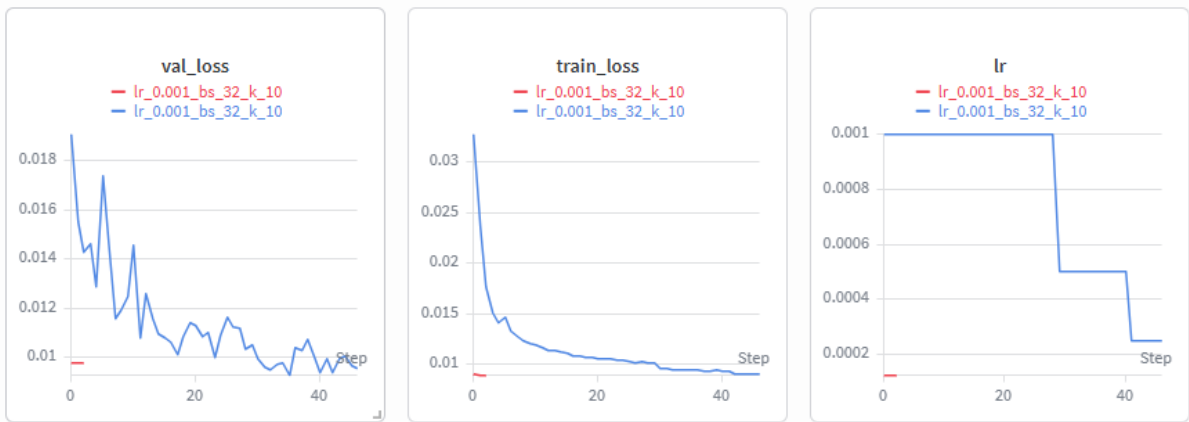
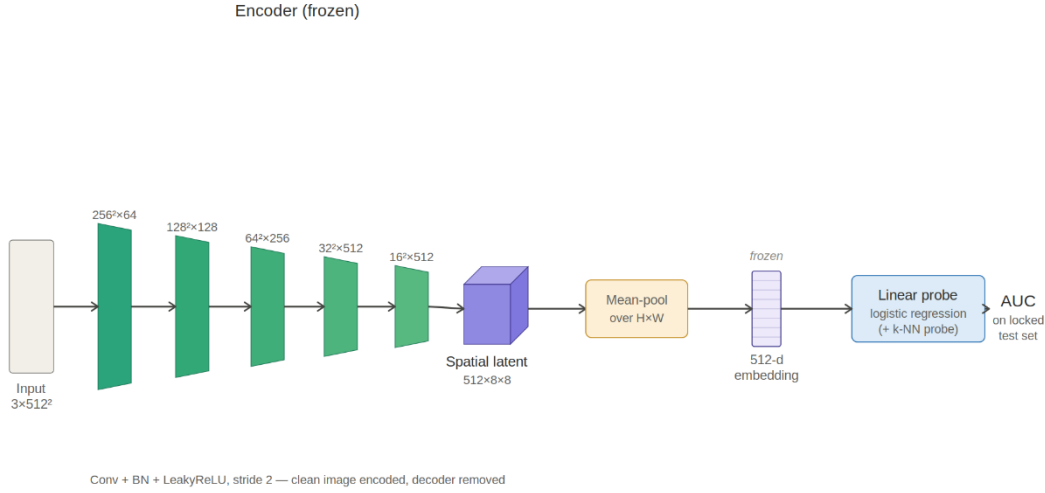


Figure A16. Training and validation reconstruction loss against epoch for the GAE. The right panel shows the learning rate over training.

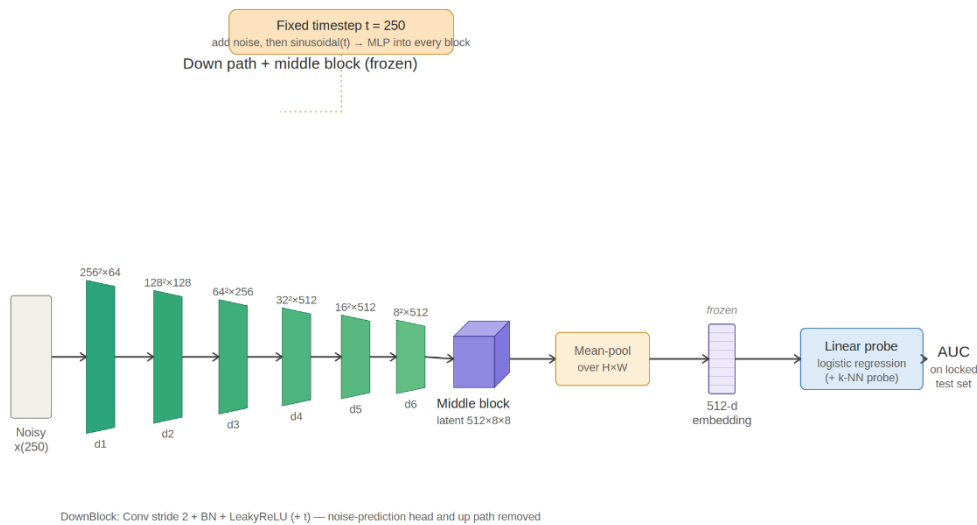
Appendix B

B1. Architecture diagram for models during evaluation



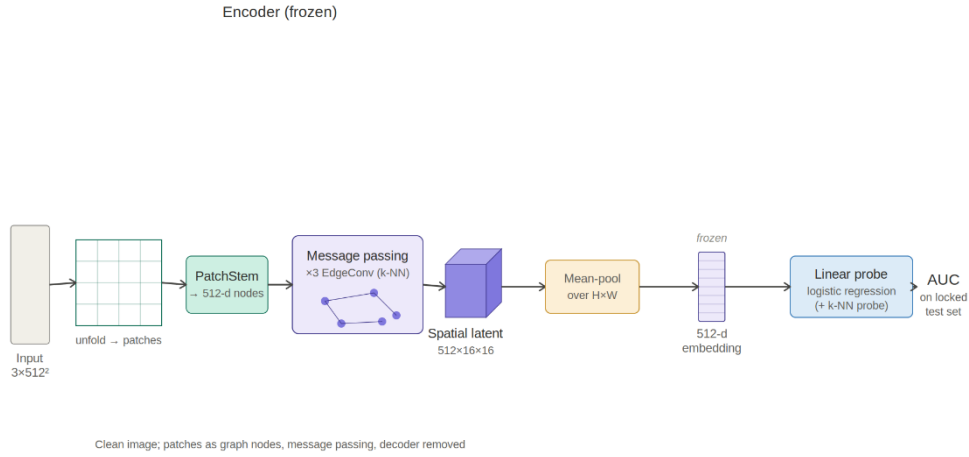
Evaluation path for the convolutional autoencoder. The decoder is removed. The clean image is passed through the frozen encoder to the $512 \times 8 \times 8$ latent, which is mean-pooled to a 512-d embedding. A logistic-regression linear probe (and a k-NN probe) is trained on the embeddings and scored by AUC on the locked test set. The encoder weights are frozen; only the probe is trained.

Figure B1. Frozen-evaluation path for the convolutional autoencoder. The decoder is removed; the clean image is passed through the frozen encoder to the $512 \times 8 \times 8$ latent, which is mean-pooled to a 512-dimensional embedding. A logistic-regression linear probe and a k-NN probe are trained on the embeddings and scored by AUC on the locked test set. Only the probe is trained.



Evaluation path for the diffusion model. The up path and noise-prediction head are removed. The image is noised once at the fixed timestep $t = 250$ and passed through the frozen down path and middle block to the $512 \times 8 \times 8$ latent, which is mean-pooled to a 512-d embedding. A logistic-regression linear probe (and a k-NN probe) is trained on the embeddings and scored by AUC on the locked test set. Encoder frozen; only the probe is trained.

Figure B2. Frozen-evaluation path for the diffusion model. The up path and noise-prediction head are removed. The image is noised once at the fixed timestep $t = 250$ and passed through the frozen down path and middle block to the $512 \times 8 \times 8$ latent, which is mean-pooled to a 512-dimensional embedding. A logistic-regression linear probe and a k -NN probe are trained on the embeddings and scored by AUC on the locked test set.



Evaluation path for the graph autoencoder. The decoder is removed. The clean image is cut into patches, encoded into graph nodes, and passed through the frozen message-passing encoder to the $512 \times 16 \times 16$ latent, which is mean-pooled to a 512-d embedding. A logistic-regression linear probe (and a k -NN probe) is trained on the embeddings and scored by AUC on the locked test set. Encoder frozen; only the probe is trained.

Figure B3. Frozen-evaluation path for the graph autoencoder. The decoder is removed; the clean image is cut into patches, encoded into graph nodes, and passed through the frozen message-passing encoder to the $512 \times 16 \times 16$ latent, which is mean-pooled to a 512-dimensional embedding. A logistic-regression linear probe and a k -NN probe are trained on the embeddings and scored by AUC on the locked test set.

B2. Architecture Diagram during Finetuning

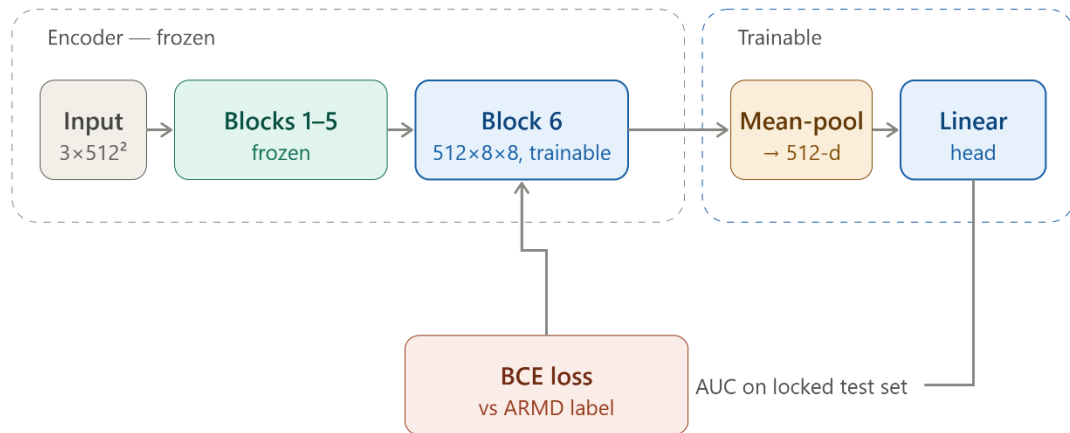


Figure B4. Fine-tuning path for the convolutional autoencoder. The clean image is passed through the encoder; the first five convolutional blocks are frozen and only the last block is trained, together with a single linear classification head on the mean-pooled 512-dimensional embedding. Training uses a binary cross-entropy loss against the ARMD label, and the model is scored by AUC on the locked test set.

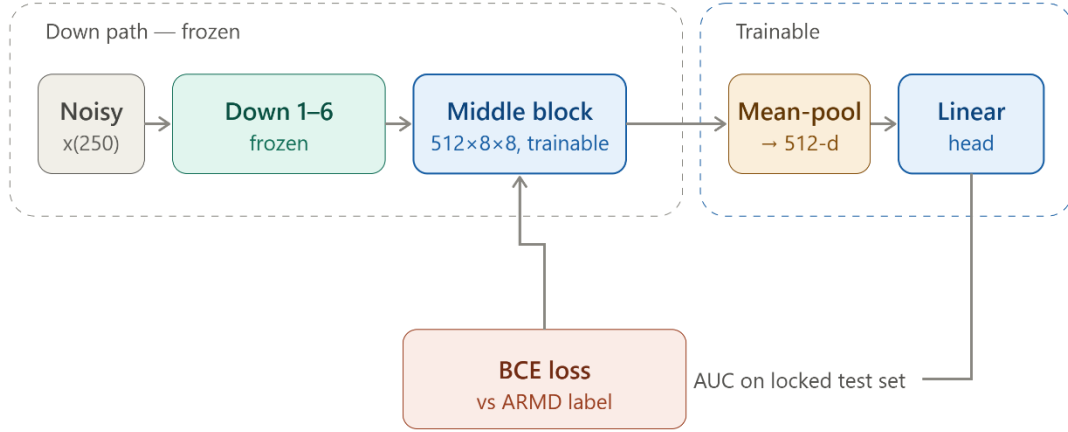


Figure B5. Fine-tuning path for the diffusion model. The image is noised at the fixed timestep $t = 250$ and passed through the down path; the down path is frozen and only the middle block is trained, together with a single linear classification head on the mean-pooled 512-dimensional embedding. Training uses a binary cross-entropy loss against the ARMD label, and the model is scored by AUC on the locked test set.

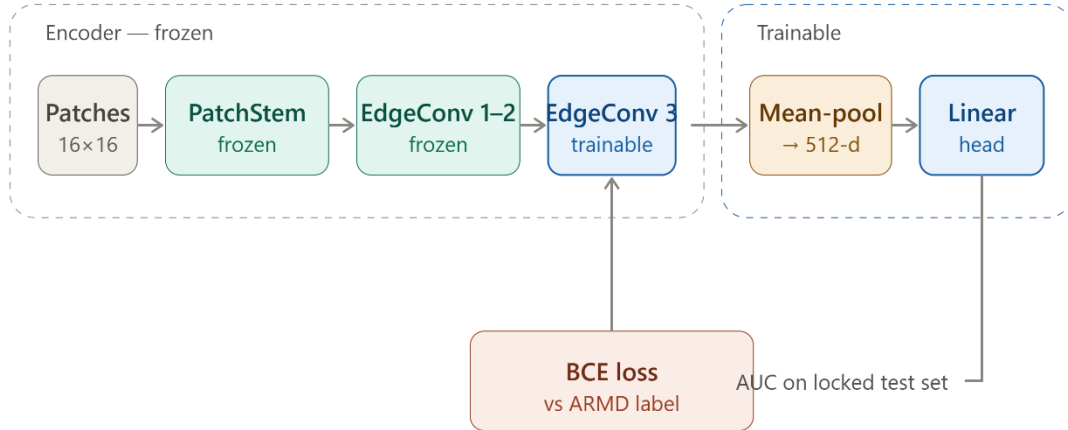


Figure B6. Fine-tuning path for the graph autoencoder. The clean image is cut into patches and passed through the message-passing encoder; the PatchStem and first two EdgeConv layers are frozen and only the final EdgeConv layer is trained, together with a single linear classification head on the mean-pooled 512-dimensional embedding. Training uses a binary cross-entropy loss against the ARMD label, and the model is scored by AUC on the locked test set.

Statement Of AI Use

- **Summarization.** AI was used to summarise reference material and source documents. All summarised content was cross-checked against the original sourced before being used.
- **Code Assistance.** AI was used as coding assistant on certain occasions during the study. All code produced this way was reviewed and understood by me, verified against the intended behaviour, and modified where required.
- **Grammar and Syntax.** AI was used for standard grammar, spelling and syntax checking.
- **Drafting Support.** AI was used on occasion to generate text as a source of ideas and phrasing. Any and all such output was treated as a starting point. The final wording and content are my own.
- **Discussion of technical Ideas.** AI was used to discuss and compare design choices for the model architectures, including topics such as compression and skip connections, over the course of the study. These discussions informed my way of thinking. But the design decisions and their implementation are my own.